

Time Series with Stationary Data

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1 Introduction

- When modeling relationships between variables, the nature of the data that have been collected has an important bearing on the appropriate choice of an econometric model.
- Two features of time-series data to consider:
 1. Time-series observations on a given economic unit, observed over a number of time periods, are likely to be correlated.
 2. Time-series data have a natural ordering according to time.
- There is also the possible existence of dynamic relationships between variables.

- A dynamic relationship is one in which the change in a variable now has an impact on that same variable, or other variables, in one or more future time periods.
- These effects do not occur instantaneously but are spread, or *distributed*, over future time periods.

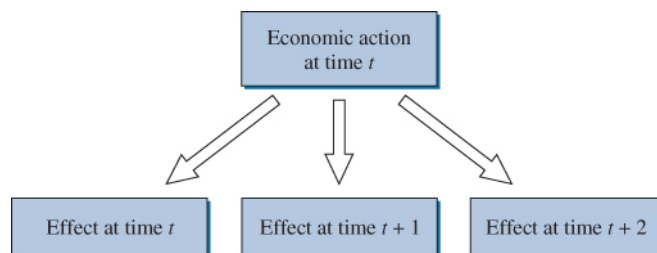


Figure 1: The Distributed Lag Effect

1.1 Ways to model the dynamic relationship–1:

- Specify that a dependent variable y is a function of current and past values of an explanatory variable x

$$y_t = f(x_t, x_{t-1}, x_{t-2}, \dots) \quad (1)$$

- Because of the existence of these lagged effects, Eq. 1 is called a distributed lag model.

1.2 Ways to model the dynamic relationship–2:

- Capturing the dynamic characteristics of time-series by specifying a model with a lagged dependent variable as one of the explanatory variables

$$y_t = f(y_{t-1}, x_t) \quad (2)$$

- Or have:

$$y_t = f(y_{t-1}, x_t, x_{t-1}, x_{t-2}) \quad (3)$$

- Such models are called *autoregressive distributed lag* (ARDL) models, with “autoregressive” meaning a regression of y_t on its own lag or lags.

1.3 Ways to model the dynamic relationship–3:

- Model the continuing impact of change over several periods via the error term

$$y_t = f(x_t) + e_{t-1} \quad (4)$$

- In this case e_t is correlated with e_{t-1} .
- We say the errors are *serially correlated* or *autocorrelated*.

1.4 Assumptions

- The primary assumption is Assumption MR4:

$$\text{Cov}(y_i, y_j) = \text{Cov}(e_i, e_j) = 0, \quad \text{for } i \neq j$$

- For time series, this is written as:

$$\text{Cov}(y_t, y_s) = \text{Cov}(e_t, e_s) = 0, \quad \text{for } t \neq s$$

- The dynamic models in Eqs. 2, 3 and 4 imply correlation between y_t and y_{t-1} or e_t and e_{t-1} or both, so they clearly violate assumption MR4.
- A stationary variable is one that is not explosive, nor trending, and nor wandering aimlessly without returning to its mean.

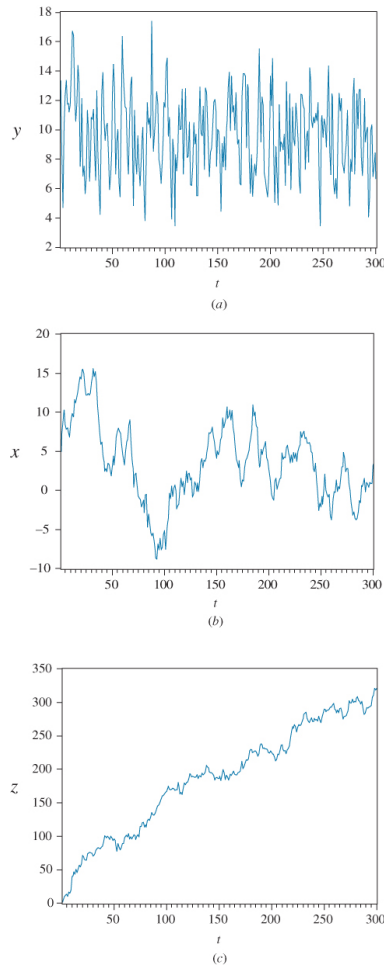


Figure 2: (a) Time series of a stationary variable (b) Time series of a nonstationary variable that is “slow-turning” or “wandering” (c) Time series of a nonstationary variable that “trends”.

2 Finite Distributed Lags

- Consider a linear model in which, after q time periods, changes in x no longer have an impact on y

$$y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \beta_2 x_{t-2} + \cdots + \beta_q x_{t-q} + e_t \quad (5)$$

- Note the notation change: β_s is used to denote the coefficient of x_{t-s} and α is introduced to denote the intercept.
- Model 5 has two uses:

Forecasting

$$y_{T+1} = \alpha + \beta_0 x_{T+1} + \beta_1 x_T + \beta_2 x_{T-1} + \cdots + \beta_q x_{T-q+1} + e_{T+1} \quad (6)$$

Policy analysis

- What is the effect of a change in x on y ?

$$\frac{\partial E(y_t)}{\partial x_{t-s}} = \frac{\partial E(y_{t+s})}{\partial x_t} = \beta_s \quad (7)$$

- Assume x_t is increased by one unit and then maintained at its new level in subsequent periods.
 - The immediate impact will be β_0 .
 - the total effect in period $t+1$ will be $\beta_0 + \beta_1$, in period $t+2$ it will be $\beta_0 + \beta_1 + \beta_2$, and so on.
 - These quantities are called *interim multipliers*.
 - The *total multiplier* is the final effect on y of the sustained increase after q or more periods have elapsed $\sum_{s=0}^q \beta_s$.
- The effect of a one-unit change in x_t is *distributed* over the current and next q periods, from which we get the term “distributed lag model”.
 - It is called a *finite distributed lag model of order q* .
 - It is assumed that after a finite number of periods q , changes in x no longer have an impact on y .
 - The coefficient β_s is called a *distributed-lag weight* or an *s-period delay multiplier*.
 - The coefficient β_0 ($s = 0$) is called the *impact multiplier*.

2.1 Assumptions of the Distributed lag model

- TSMR1.** $y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \beta_2 x_{t-2} + \cdots + \beta_q x_{t-q} + e_t, t = q+1, \dots, T$.
- TSMR2.** y and x are stationary random variables, and e_t is independent of current, past and future values of x .
- TSMR3.** $E(e_t) = 0$
- TSMR4.** $Var(e_t) = \sigma^2$
- TSMR5.** $Cov(e_t, e_s) = 0$, for $t \neq s$
- TSMR6.** $e_t \sim N(0, \sigma^2)$

Example: Consider Okun's Law (relates losses in production to unemployment),

- In this model the change in the unemployment rate from one period to the next depends on the rate of growth of output in the economy:

$$U_t - U_{t-1} = -\gamma (G_t - G_N) \quad (8)$$

- We can rewrite this as:

$$DU_t = \alpha + \beta_0 G_t + e_t \quad (9)$$

where $DU_t = \Delta U_t = U_t - U_{t-1}$, $\beta_0 = -\gamma$, and $\alpha = \gamma G_N$

- We can expand this to include lags:

$$DU_t = \alpha + \beta_0 G_t + \beta_1 G_{t-1} + \cdots + \beta_q G_{t-q} + e_t \quad (10)$$

- We can calculate the growth in output, G_t , as:

$$G_t = 100 \left(\frac{GDP_t - GDP_{t-1}}{GDP_{t-1}} \right) \quad (11)$$

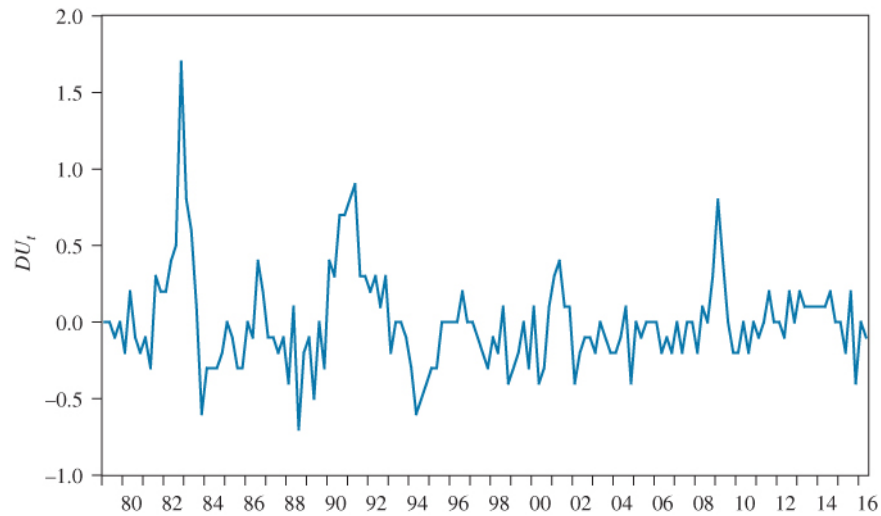


Figure 3: Time series for the change in the U.S. unemployment rate: 1978-Q3 to 2016-Q1.

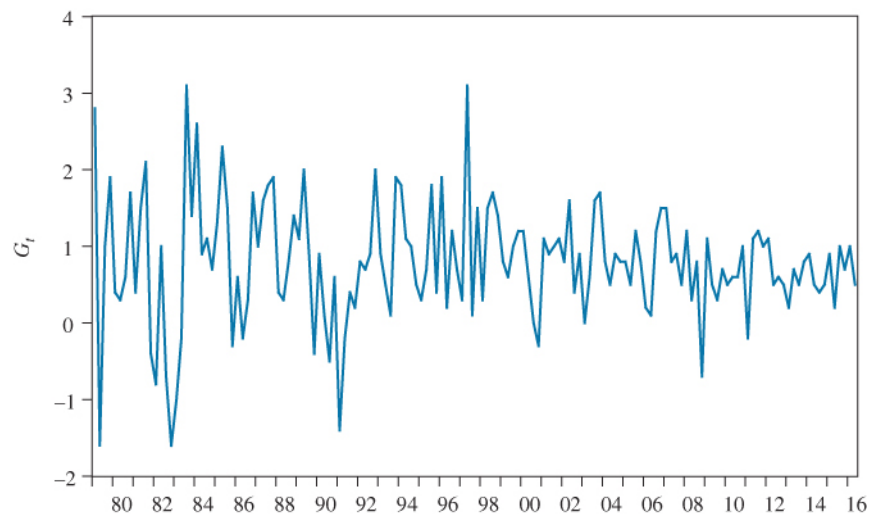
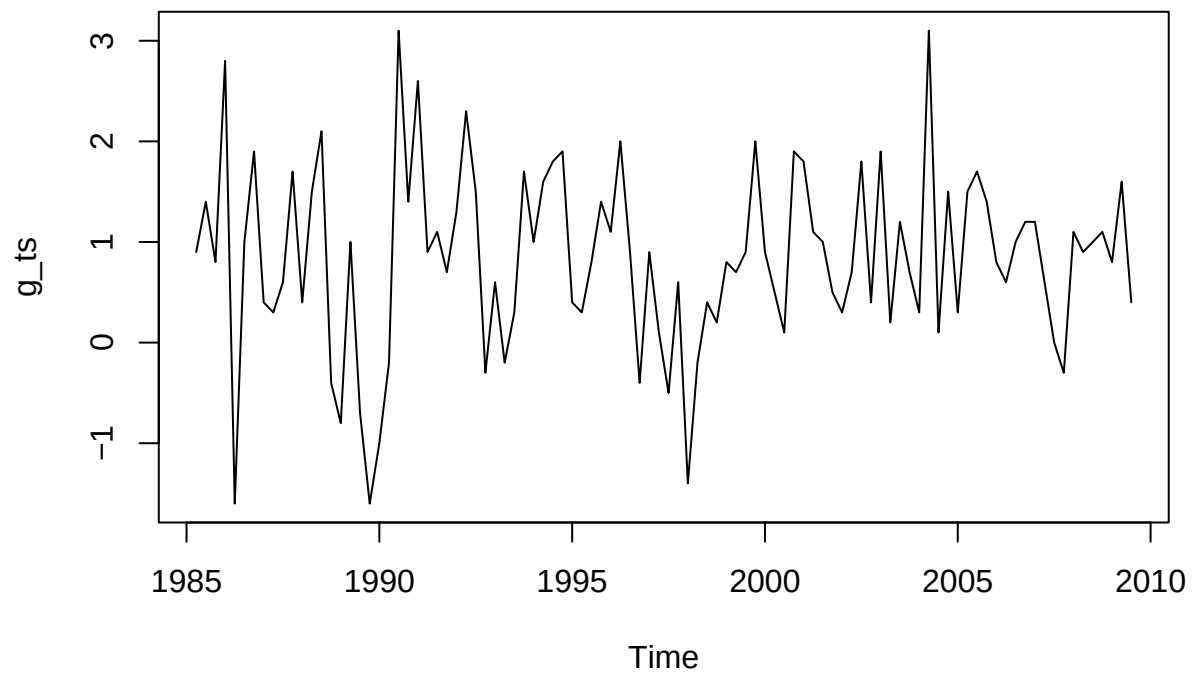


Figure 4: Time series for U.S. GDP growth: 1978-Q3 to 2016-Q1.

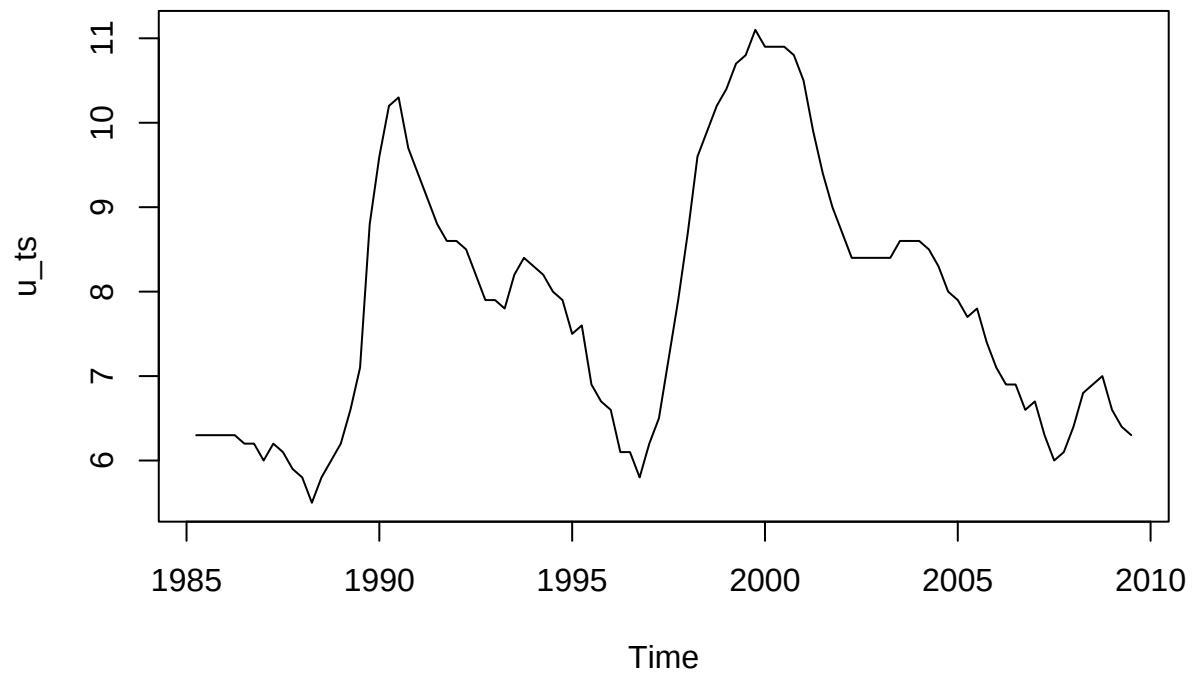
R Example

```
# Okun: quarterly (1985Q2 - 2009Q3) observations
# g = percentage change in U.S. Gross Domestic Product, seasonally adjusted
# u = U.S. Civilian Unemployment Rate (Seasonally adjusted)
library(POE5Rdata)
library(tseries)
library(forecast)
data("okun5_aus")
attach(okun5_aus)

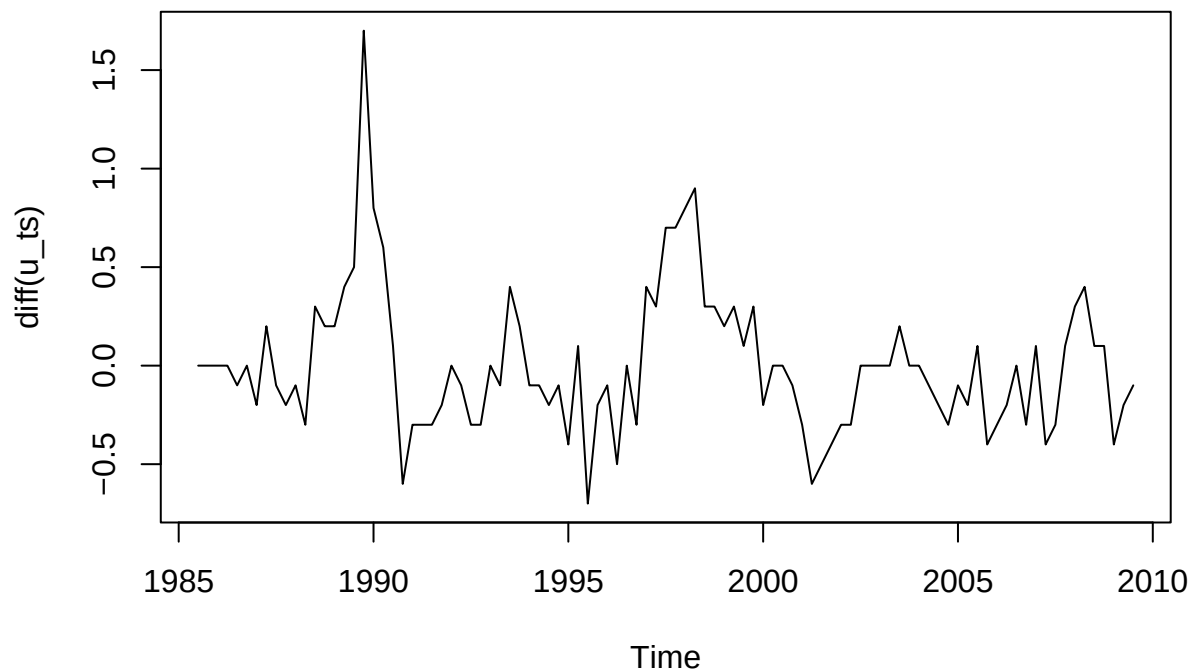
# Convert to time series format
g_ts = ts(g,start=c(1985,2), end=c(2009,3), frequency=4)
u_ts = ts(u,start=c(1985,2), end=c(2009,3), frequency=4)
plot(g_ts)
```



```
plot(u_ts)
```



```
plot(diff(u_ts))
```



```
# Regress DU_t on present and past growth rates G_{t}, G_{t-1}, ..., G_{t-5}
library(dynlm)
reg.ts = dynlm(d(u_ts)~L(g_ts,0:5))

summary(reg.ts)
```

```
##
## Time series regression with "ts" data:
## Start = 1986(3), End = 2009(3)
##
## Call:
## dynlm(formula = d(u_ts) ~ L(g_ts, 0:5))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.70139 -0.14699 -0.00117  0.14646  0.85911
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.44470    0.05558   8.001 5.26e-12 ***
## L(g_ts, 0:5)0 -0.14685    0.03008  -4.883 4.78e-06 ***
## L(g_ts, 0:5)1 -0.16874    0.02913  -5.792 1.12e-07 ***
## L(g_ts, 0:5)2 -0.10201    0.02813  -3.626 0.000487 ***
## L(g_ts, 0:5)3 -0.07346    0.02813  -2.611 0.010648 *
## L(g_ts, 0:5)4 -0.07631    0.02817  -2.709 0.008141 **
## L(g_ts, 0:5)5  0.02658    0.02796   0.951 0.344437
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2459 on 86 degrees of freedom
## Multiple R-squared:  0.5828, Adjusted R-squared:  0.5537
## F-statistic: 20.02 on 6 and 86 DF,  p-value: 1.642e-14
```


3 Serial Correlation

- When is assumption TSMR5, $Cov(e_t, e_s) = 0$ for $t \neq s$ likely to be violated, and how do we assess its validity?
- When a variable exhibits correlation over time, we say it is *autocorrelated* or *serially correlated*.
- These terms are used interchangeably.

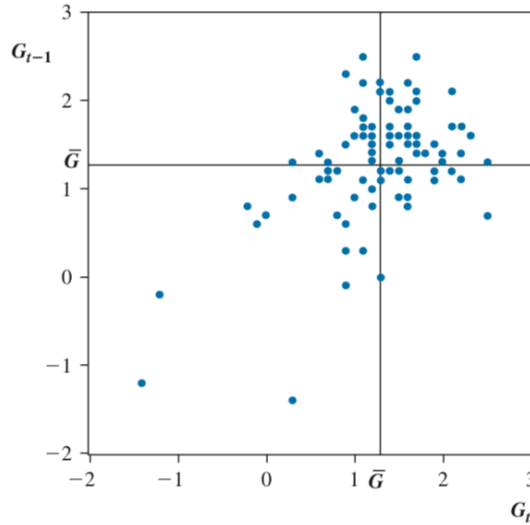


Figure 5: Scatter Diagram for G_t and G_{t-1} .

- Recall that the population correlation between two variables x and y is given by:

$$\rho_{xy} = \frac{Cov(x, y)}{\sqrt{Var(x) Var(y)}}$$

Example: Consider Okun's Law (continued)

- For the Okun's Law problem, we have:

$$\rho_1 = \frac{Cov(G_t, G_{t-1})}{\sqrt{Var(G_t) Var(G_{t-1})}} = \frac{Cov(G_t, G_{t-1})}{Var(G_t)} \quad (12)$$

- The notation ρ_1 is used to denote the population correlation between observations that are one period apart in time.
 - This is known also as the *population autocorrelation of order one*.
 - The second equality in Eq. 12 holds because $Var(G_t) = Var(G_{t-1})$, a property of time series that are stationary.
- The first-order sample autocorrelation for G is obtained from Eq. 12 using the estimates:

$$\begin{aligned} \hat{Cov}(G_t, G_{t-1}) &= \frac{1}{T-1} \sum_{t=2}^T (G_t - \bar{G})(G_{t-1} - \bar{G}) \\ \hat{Var}(G_t) &= \frac{1}{T-1} \sum_{t=2}^T (G_t - \bar{G})^2 \end{aligned}$$

- Making the substitutions, we get:

$$r_1 = \frac{\sum_{t=2}^T (G_t - \bar{G}) (G_{t-1} - \bar{G})}{\sum_{t=2}^T (G_t - \bar{G})^2} \quad (13)$$

- More generally, the ***k*-th order sample autocorrelation** for a series y that gives the correlation between observations that are k periods apart is:

$$r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y}) (y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2} \quad (14)$$

- Because $(T - k)$ observations are used to compute the numerator and T observations are used to compute the denominator, an alternative that leads to larger estimates in finite samples is:

$$r'_k = \frac{\frac{1}{T-k} \sum_{t=k+1}^T (y_t - \bar{y}) (y_{t-k} - \bar{y})}{\frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2} \quad (15)$$

- Applying this to our problem, we get for the first four autocorrelations:

$$r_1 = 0.494, \quad r_2 = 0.411, \quad r_3 = 0.154, \quad r_4 = 0.200 \quad (16)$$

- How do we test whether an autocorrelation is significantly different from zero?
- The null hypothesis is $H_0: \rho_k = 0$
- A suitable test statistic is:

$$Z = \frac{r_k - 0}{\sqrt{1/T}} \sim N(0, 1) \quad (17)$$

- For our problem, we have:

$$\begin{aligned} Z_1 &= \sqrt{98} \times 0.494 = 4.89, & Z_2 &= \sqrt{98} \times 0.411 = 4.10 \\ Z_3 &= \sqrt{98} \times 0.154 = 1.52, & Z_4 &= \sqrt{98} \times 0.200 = 1.98 \end{aligned}$$

- Using $Z_c = \pm 1.96$, we reject the hypotheses $H_0: \rho_k = 0$ when $\sqrt{T}r_k \geq 1.96$ or $\sqrt{T}r_k \leq -1.96$.
- Based on our estimates: we have insufficient evidence to reject $H_0: \rho_3 = 0$, however, we do reject $H_0: \rho_1 = 0$, and $\rho_2 = 0$.
- ρ_4 is on the borderline of being significant
- We conclude that G , the quarterly growth rate in U.S. GDP, exhibits significant serial correlation at lags one and two.
- The *correlogram*, also called the *sample autocorrelation function*, is the sequence of autocorrelations $r_1, r_2, r_3, \dots$
- It shows the correlation between observations that are one period apart, two periods apart, three periods apart, and so on.

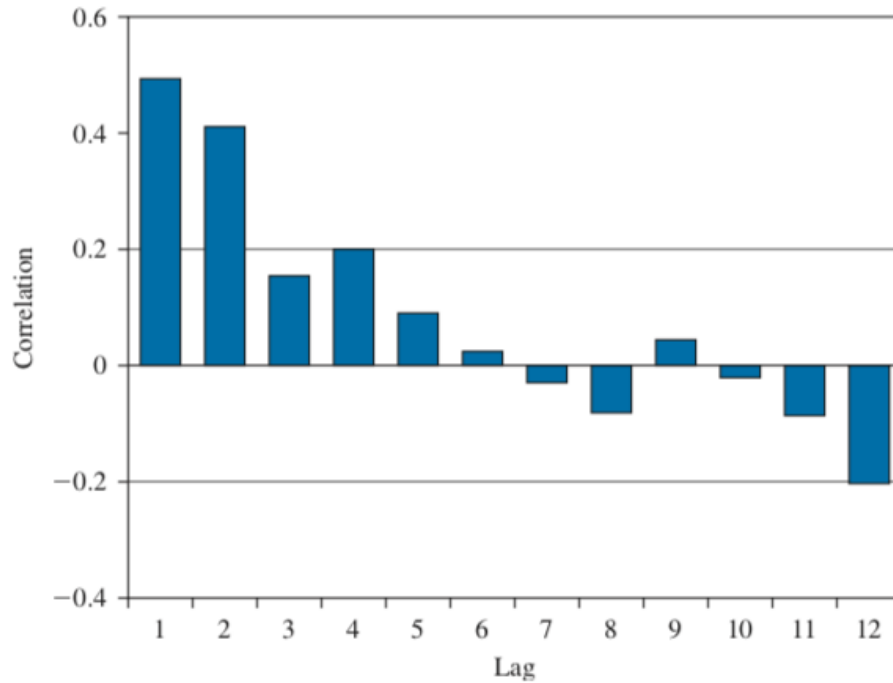


Figure 6: Correlogram for G.

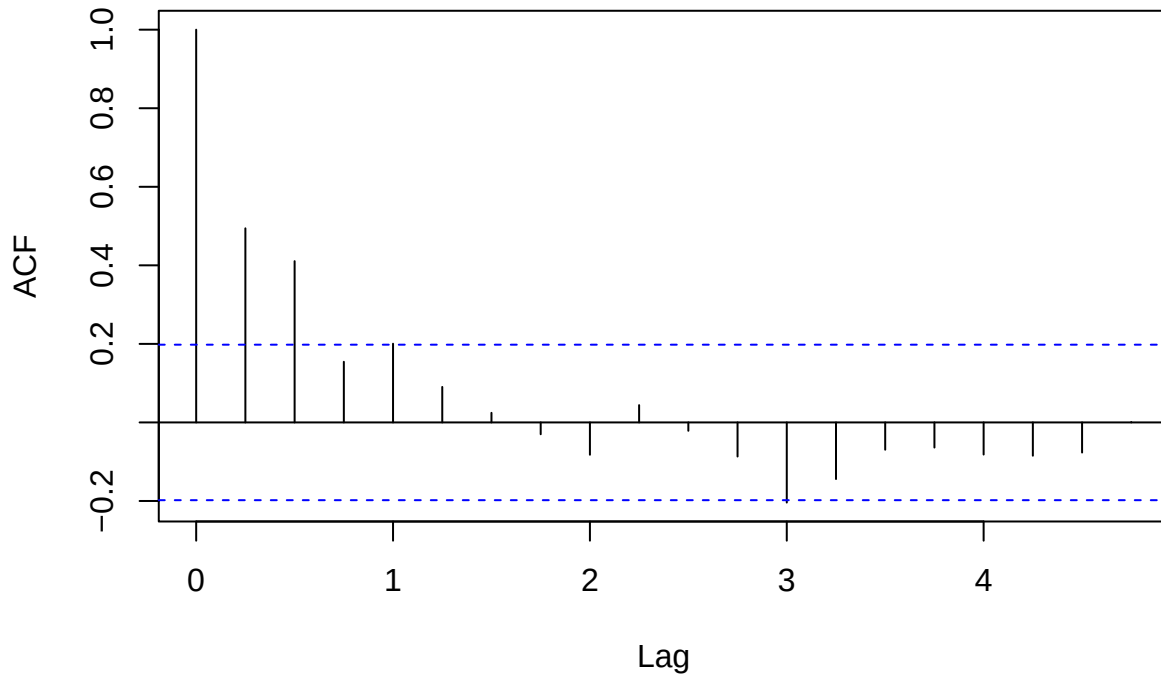
R Example

```
# Okun: quarterly (1985Q2 - 2009Q3) observations
# g = percentage change in U.S. Gross Domestic Product, seasonally adjusted
# u = U.S. Civilian Unemployment Rate (Seasonally adjusted)

library(POE5Rdata)
library(tseries)
library(forecast)
data("okun5_aus")
attach(okun5_aus)
rm(g_ts)
# Convert to time series format
g_ts = ts(g,start=c(1985,2), end=c(2009,3), frequency=4)
u_ts = ts(u,start=c(1985,2), end=c(2009,3), frequency=4)
library(stats)

library(PoEdata)
data(okun)
okun.ts <- ts(okun, start=c(1985,2), end=c(2009,3),frequency=4)
growth_rate <- okun.ts[, "g"]
acf(growth_rate)
```

Series growth_rate



- The correlogram can also be used to check whether the multiple regression assumption $Cov(e_t, e_s) = 0$ for $t \neq s$ is violated.

Example: Consider a model for a Phillips Curve,

$$INF_t = INF_t^E - \gamma (U_t - U_{t-1}) \quad (18)$$

- If we initially assume that inflationary expectations are constant over time ($\beta_1 = INF_t^E$), set $\beta_2 = -\gamma$, and add an error term:

$$INF_t = \beta_1 + \beta_2 DU_t + e_t \quad (19)$$

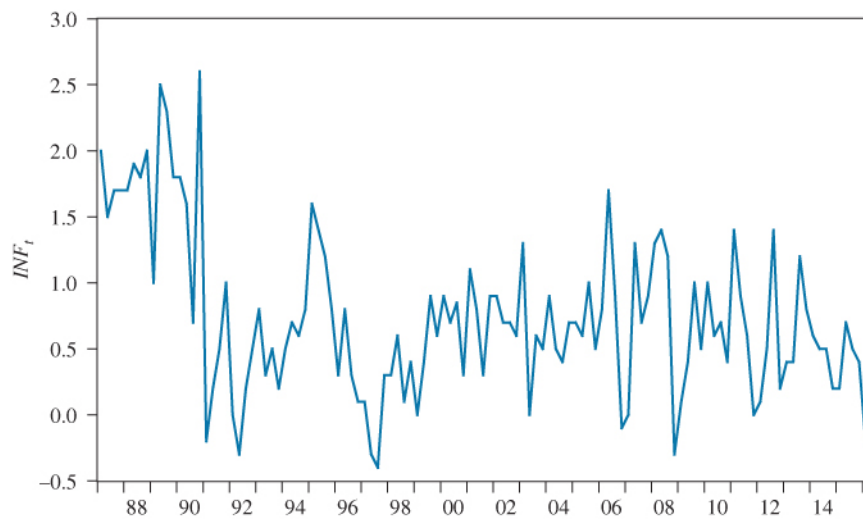


Figure 7: Time series for Australian Price Inflation

- To determine if the errors are serially correlated, we compute the least squares residuals:

$$\hat{e}_t = INF_t - b_1 - b_2 DU_t \quad (20)$$

- The k -th order autocorrelation for the residuals can be written as:

$$r_k = \frac{\sum_{t=k+1}^T \hat{e}_t \hat{e}_{t-k}}{\sum_{t=k+1}^T \hat{e}_t^2} \quad (21)$$

- The least squares equation is:

$$\begin{array}{rcl} \widehat{INF} = & 0.7776 & -0.5279DU \\ (se) & (0.0658) & (0.2294) \end{array} \quad (22)$$

- The autocorrelation values at the first five lags are:

$$\begin{array}{rcl} r_1 & = & 0.549, \quad r_2 = 0.456, \quad r_3 = 0.433, \\ r_4 & = & 0.420, \quad r_5 = 0.339. \end{array}$$

These autocorrelation values seem large in magnitude, suggesting that our model hasn't fully captured all the respective dynamics.

R Example

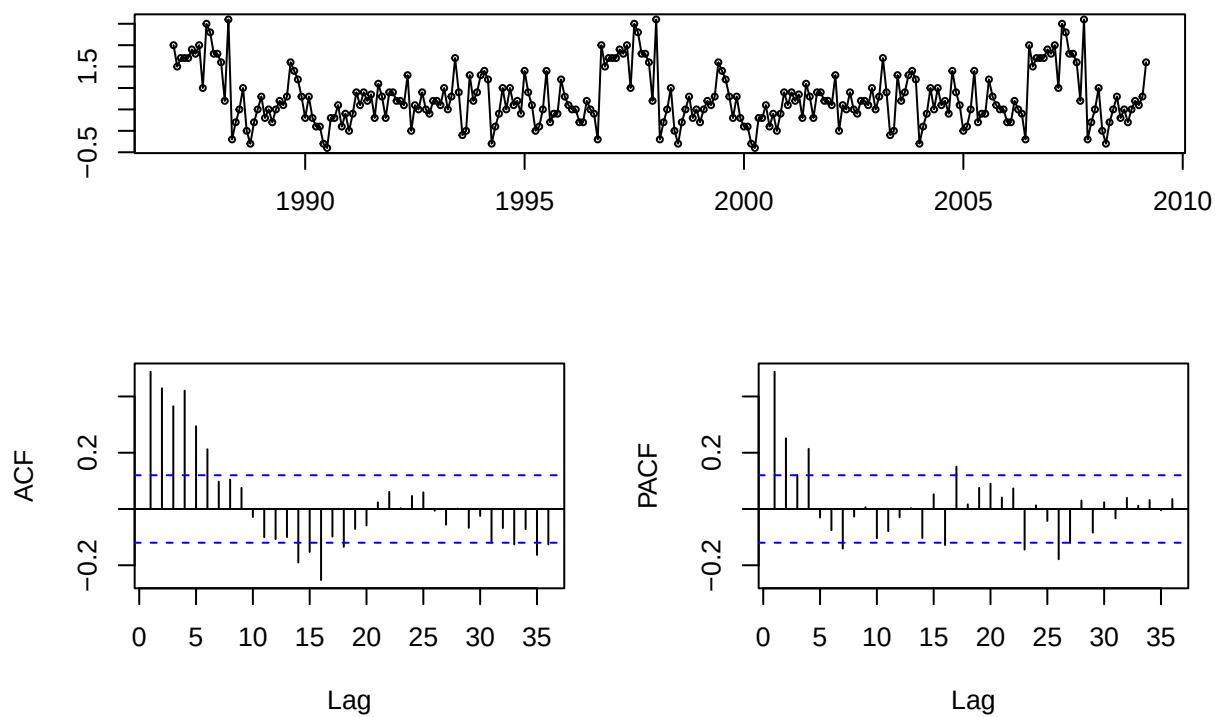
```
library(POE5Rdata)
library(tseries)
library(forecast)
data("phillips5_aus")
attach(phillips5_aus)

# phillips: monthly (1987:01 - 2009:3) observations on
#   inf = Australian Inflation Rate (Percentage Change in the Consumer Price Index
#         with an adjustment for 2000Q3 when a national sales tax was introduced)
#   u = Australian Unemployment Rate (Seasonally adjusted)

phillips5_aus.ts <- ts(phillips5_aus,
                      start=c(1987,1),
                      end=c(2009,3),
                      frequency=12)
inf.ts <- phillips5_aus.ts[, "inf"]
Du.ts <- diff(phillips5_aus.ts[, "u"])

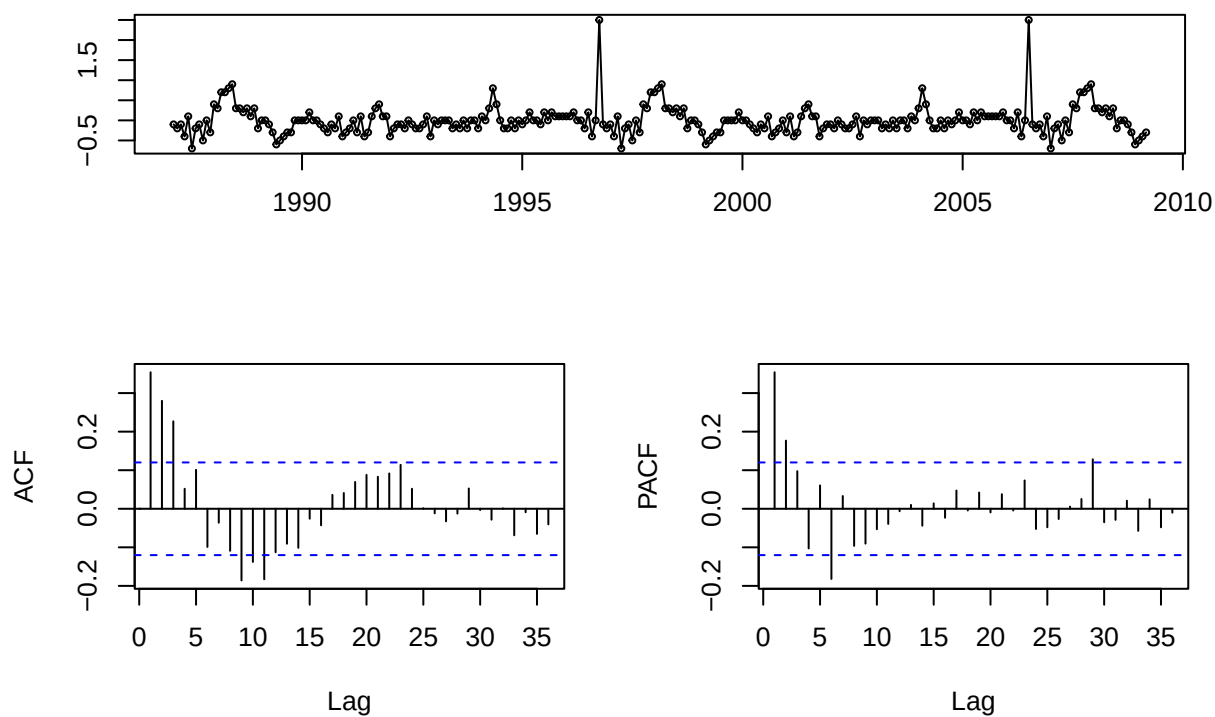
tsdisplay(inf.ts)
```

inf.ts



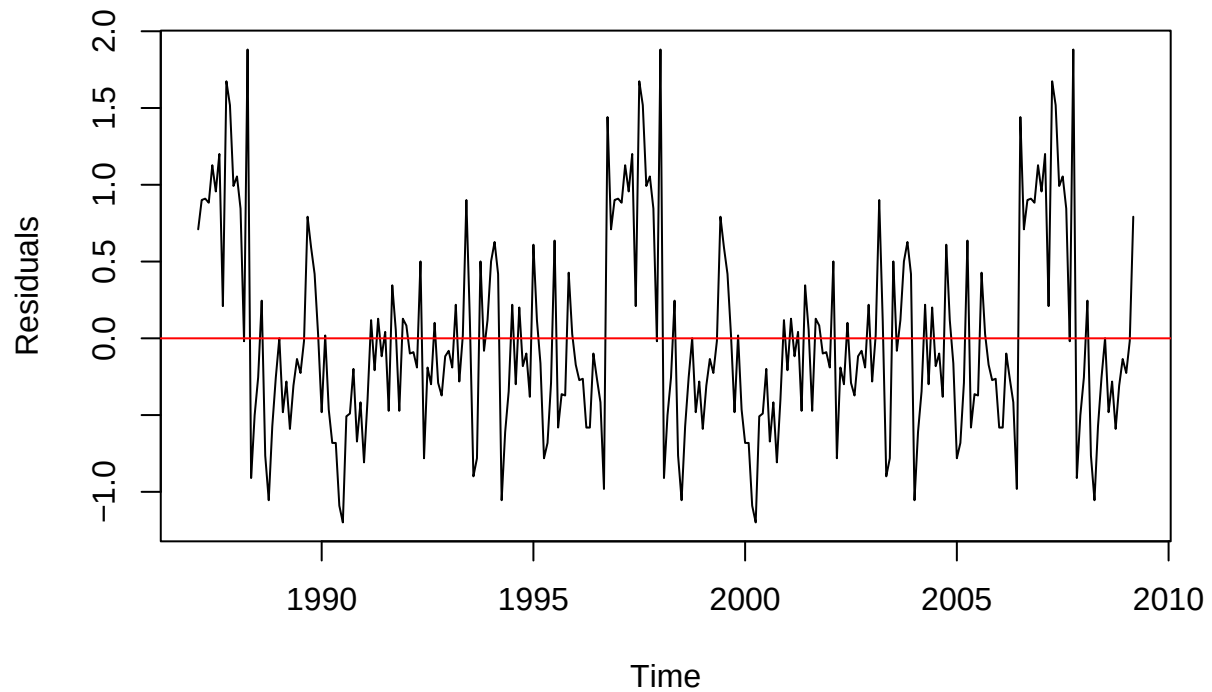
```
tsdisplay(Du.ts)
```

Du.ts



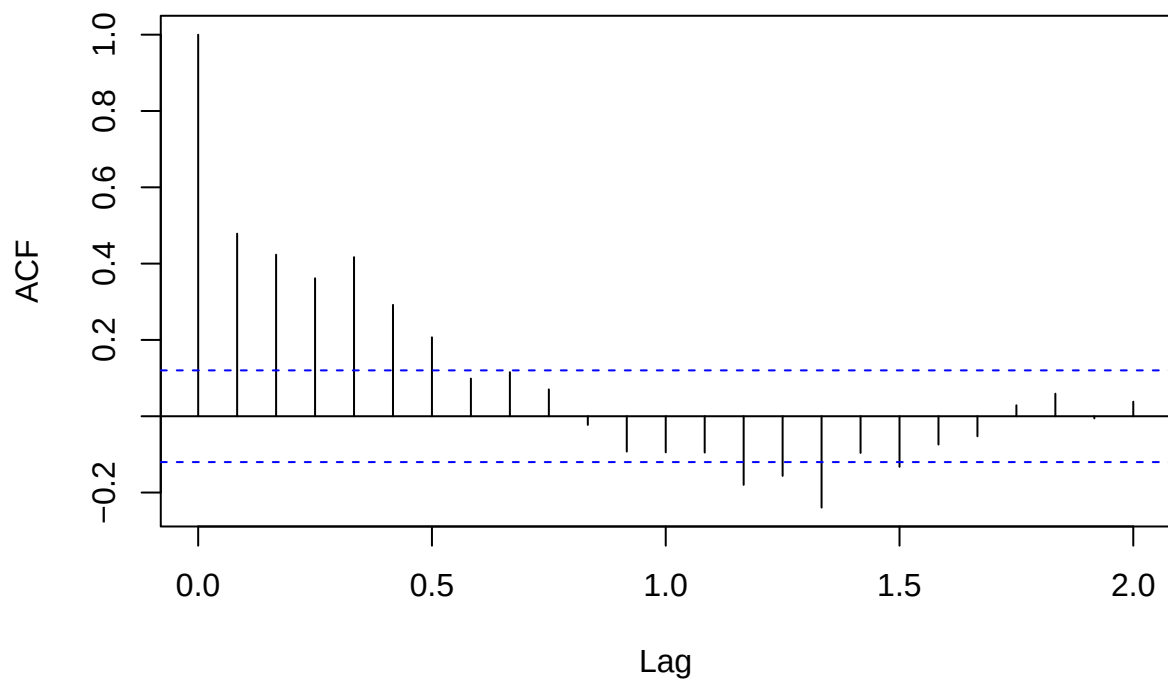
```
# Regress inflation of unemployment rate
reg.ts = dynlm(inf.ts~Du.ts,data=phillips5_au.ts)
```

```
plot(reg.ts$residuals, pch=20, ylab="Residuals")
abline(h=0, lwd=1, col="red")
```



```
acf(reg.ts$residuals, main="ACF of thge Residuals")
```

ACF of thge Residuals



```
# The spikes at lags <= 2 show serial correlation
# This suggests the model may not be as appropriate
```

```
# since there are dynamics unaccounted for
#print(acf(reg.ts$residuals)$acf)
```

3.1 Other Tests for Serially Correlated Errors

3.1.1 A Lagrange Multiplier Test

- An advantage of e.g., the Lagrange Multiplier (LM) Test is that it readily generalizes to a joint test of correlations at more than one lag.
- If e_t and e_{t-1} are correlated, then one way to model the relationship between them is to write:

$$e_t = \rho e_{t-1} + v_t \quad (23)$$

- We can substitute this into a simple regression equation:

$$y_t = \beta_1 + \beta_2 x_t + \rho e_{t-1} + v_t \quad (24)$$

- We have one complication: $\hat{e}_0$ is unknown
- Two ways to handle this are:
 - (i) Delete the first observation and use a total of T observations
 - (ii) Set $\hat{e}_0$ and use all T observations

Example: Model for a Phillips Curve (continued)

$$\begin{aligned} \text{(i)} \quad t &= 6.219, \quad F = 38.67, \quad p\text{-value} = 0.000 \\ \text{(ii)} \quad t &= 6.202, \quad F = 38.47, \quad p\text{-value} = 0.000 \end{aligned}$$

- The results are almost identical.
- The null hypothesis $H_0: \rho = 0$ is rejected at all conventional significance.
- We conclude that the errors are serially correlated.
- To derive the relevant auxiliary regression for the autocorrelation LM test, we write the test equation as:

$$y_t = \beta_1 + \beta_2 x_t + \rho \hat{e}_{t-1} + v_t \quad (25)$$

- But since we know that $y_t = b_1 + b_2 x_t + \hat{e}_t$, we get:

$$y_t = b_1 + b_2 x_t + \hat{e}_t = \beta_1 + \beta_2 x_t + \rho \hat{e}_{t-1} + v_t$$

- Rearranging, we get:

$$\begin{aligned} \hat{e}_t &= (\beta_1 - b_1) + (\beta_2 - b_2) x_t + \rho \hat{e}_{t-1} + v_t \\ &= \gamma_1 + \gamma_2 x_t + \rho \hat{e}_{t-1} + v_t \end{aligned} \quad (26)$$

- If $H_0: \rho = 0$ is true, then $LM = T \times R^2$ has an approximate $\chi^2_{(1)}$ distribution.
- T and R^2 are the sample size and goodness-of-fit statistic, respectively, from least squares estimation of Eq. 26.

- Considering the two alternative ways to handle $\hat{e}_0$:

$$(iii) \quad LM = (T - 1) \times R^2 = 89 \times 0.3102 = 27.61$$

$$(iv) \quad LM = T \times R^2 = 90 \times 0.3066 = 27.59$$

- These values are much larger than 3.84, which is the 5% critical value from a $\chi^2_{(1)}$ -distribution.
- We reject the null hypothesis of no autocorrelation.
- Alternatively, we can reject H_0 by examining the p -value for $LM = 27.61$, which is 0.000.
- For a four-period lag, we obtain:

$$(iii) \quad LM = (T - 4) \times R^2 = 86 \times 0.3882 = 33.4$$

$$(iv) \quad LM = T \times R^2 = 90 \times 0.4075 = 36.7$$

- Because the 5% critical value from a $\chi^2_{(4)}$ -distribution is 9.49, these LM values lead us to conclude that the errors are serially correlated.

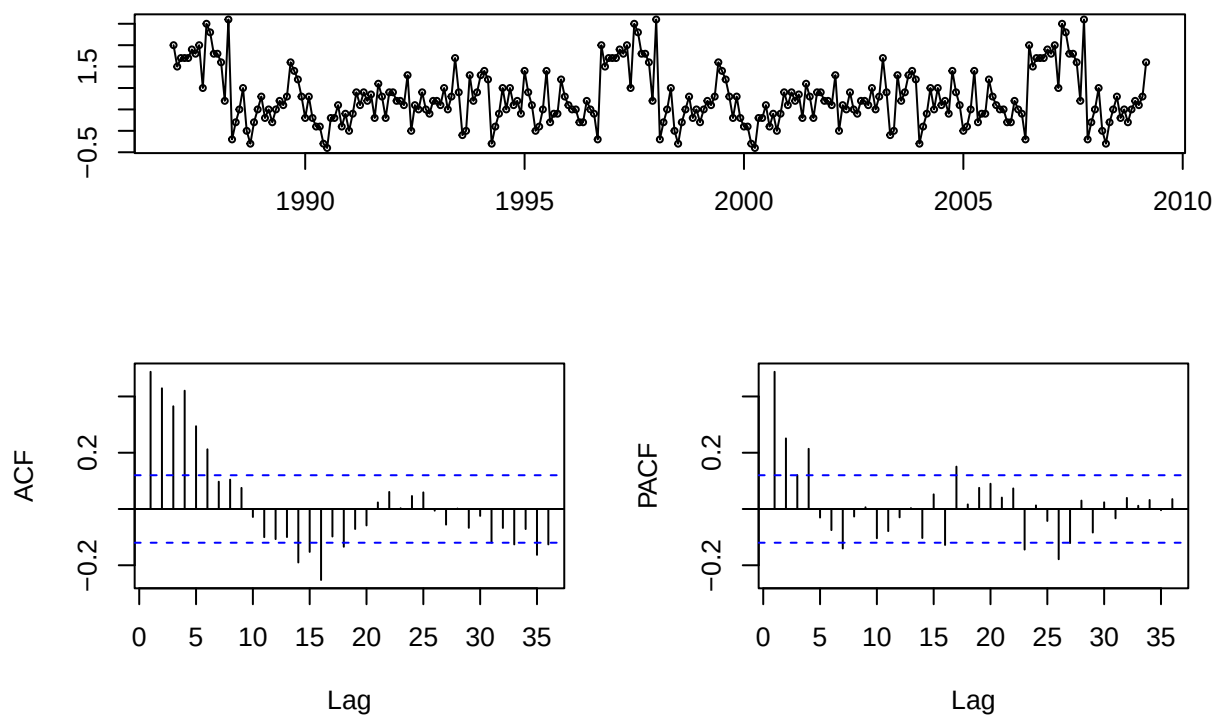
R Example

```
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#   inf = Australian Inflation Rate (Percentage Change in the Consumer Price Index
#         with an adjustment for 2000Q3 when a national sales tax was introduced)
#   u = Australian Unemployment Rate (Seasonally adjusted)
library(POE5Rdata)
library(tseries)
library(forecast)
data("phillips5_au")
attach(phillips5_au)
library(dynlm)

phillips_au.ts <- ts(phillips5_au,
                    start=c(1987,1),
                    end=c(2009,3),
                    frequency=12)
inf.ts <- phillips_au.ts[, "inf"]
Du.ts <- diff(phillips_au.ts[, "u"])

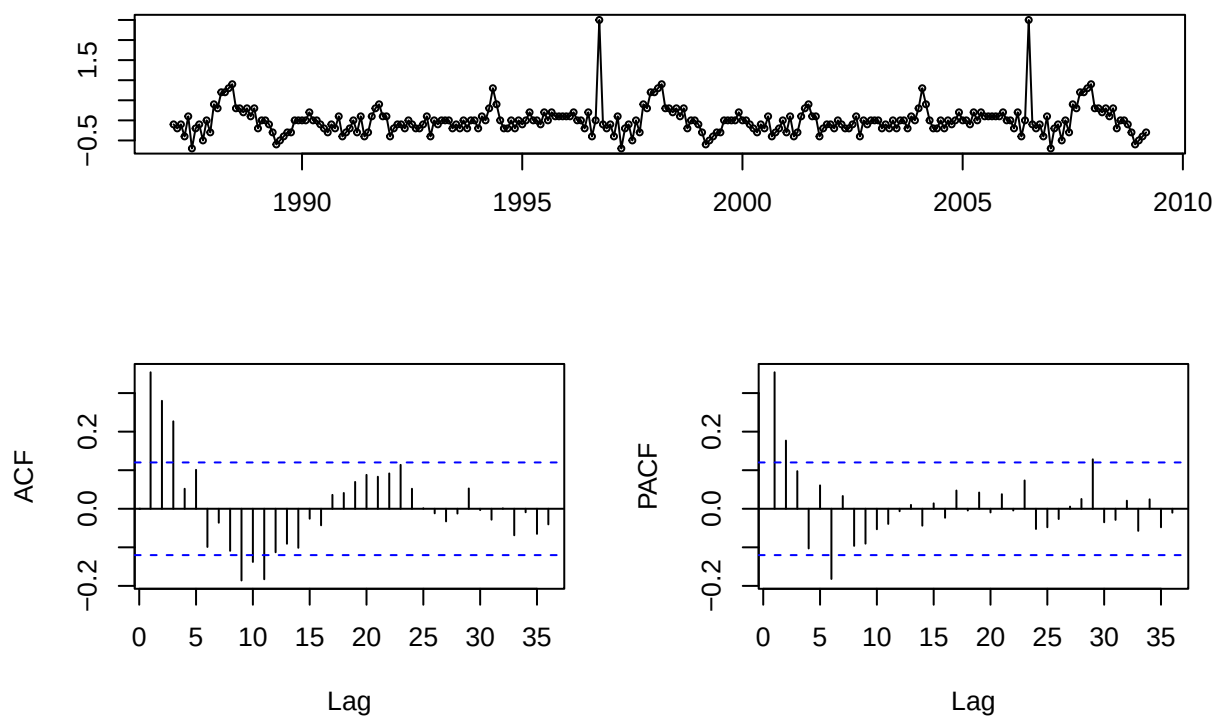
tsdisplay(inf.ts)
```

inf.ts



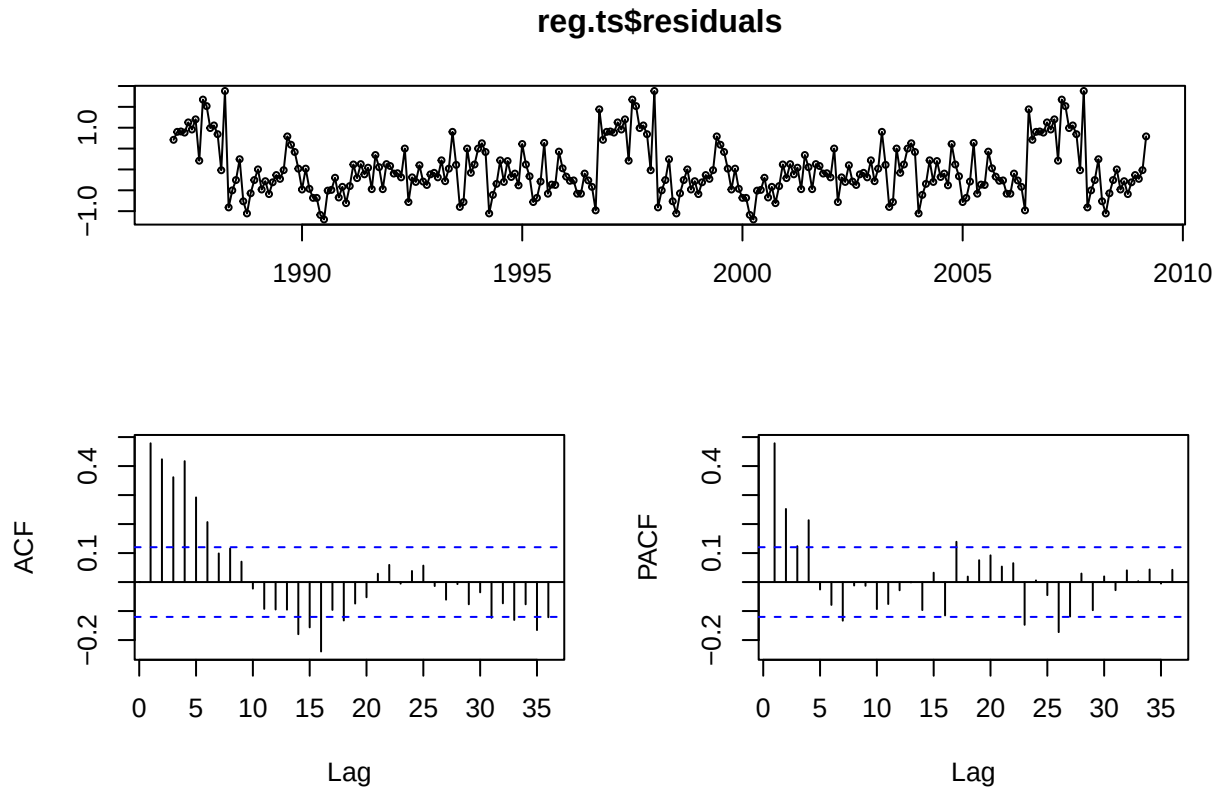
```
tsdisplay(Du.ts)
```

Du.ts



```
# Regress inflation of unemployment rate
reg.ts = dynlm(inf.ts~Du.ts,data=phillips5_au.ts)
```

```
tsdisplay(reg.ts$residuals)
```



```
# To test for serial correlation we can perform the
# "Breusch-Godfrey" test for higher-order serial correlation
library(lmtest)
```

```
# Method 1: set e_0 = 0
bgtest(reg.ts, order=1, type="F", fill=0)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data: reg.ts
## LM test = 79.398, df1 = 1, df2 = 263, p-value < 2.2e-16
```

```
# Method 2: delete the first observation
bgtest(reg.ts, order=1, type="F", fill=NA)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data: reg.ts
## LM test = 79.611, df1 = 1, df2 = 262, p-value < 2.2e-16
```

```
# In both cases we conclude that the serial correlation exists
```

4 Estimation with Serially Correlated Errors

- Three estimation procedures are considered:
 1. Least squares estimation.
 2. An estimation procedure that is relevant when the errors are assumed to follow what is known as a first-order autoregressive model (AR(1)): $e_t = \rho e_{t-1} + v_t$.
 3. A general estimation strategy for estimating models with serially correlated errors.
- We need to introduce one more assumption since we will encounter models with a lagged dependent variable, such as:

$$y_t = \delta + \theta_1 y_{t-1} + \delta_0 x_t + \delta_1 x_{t-1} + v_t$$

Assumption of the Distributed Lag Model

- **TSMR2A.** In the multiple regression model $y_t = \beta_1 + \beta_2 x_{t2} + \dots + \beta_K x_{tK} + v_t$, where some of the x_{tk} may be lagged values of y , v_t is uncorrelated with all x_{tk} and their past values.

4.1 Least Squares Estimation

- Suppose we proceed with least squares estimation without recognizing the existence of serially correlated errors. What are the consequences?
 1. The least squares estimator is still a linear unbiased estimator, but it is no longer best.
 2. The formulas for the standard errors usually computed for the least squares estimator are no longer correct.

Confidence intervals and hypothesis tests that use these standard errors may be misleading.

- It is possible to compute correct standard errors for the least squares estimator:
HAC (heteroskedasticity and autocorrelation consistent) standard errors, or Newey-West standard errors.
- These are analogous to the heteroskedasticity consistent standard errors.
- Consider the model

$$y_t = \beta_1 + \beta_2 x_t + e_t$$

- The variance of b_2 is:

$$\begin{aligned} \text{Var}(b_2) &= \sum_t w_t^2 \text{Var}(e_t) + \sum_t \sum_{s \neq t} w_t w_s \text{Cov}(e_t, e_s) \\ &= \sum_t w_t^2 \text{Var}(e_t) \left[1 + \frac{\sum_t \sum_{s \neq t} w_t w_s \text{Cov}(e_t, e_s)}{\sum_t w_t^2 \text{Var}(e_t)} \right] \end{aligned} \quad (27)$$

where

$$w_t = \frac{(x_t - \bar{x})}{\sum_t (x_t - \bar{x})^2}$$

- When the errors are not correlated, $\text{Cov}(e_t, e_s) = 0$, and the term in square brackets is equal to one.
- The resulting expression

$$\text{Var}(b_2) = \sum_t w_t^2 \text{Var}(e_t)$$

is the one used to find heteroskedasticity-consistent (HC) standard errors.

- When the errors are correlated, the term in square brackets is estimated to obtain HAC standard errors.

- If we call the quantity in square brackets g and its estimate $\hat{g}$, then the relationship between the two estimated variances is:

$$\widehat{Var}_{HAC}(b_2) = \widehat{Var}_{HC}(b_2) \times \hat{g} \quad (28)$$

Example: Let's reconsider the Phillips Curve model,

$$\widehat{INF} = 0.7776 - 0.5279DU \quad (29)$$

(0.0658)	(0.2294)	Incorrect <i>se</i>
(0.1030)	(0.3127)	HAC <i>se</i>

- The t and p -values for testing $H_0: \beta_2 = 0$ are:

$$t = \frac{-0.5279}{0.2294} = -2.301,$$

p -value = 0.0238 from LS standard errors

$$t = \frac{-0.5279}{0.3127} = -1.688,$$

p -value = 0.0950 from HAC standard errors

4.2 Estimating an AR(1) Error Model

- Recall the Lagrange Multiplier test for serially correlated errors where we used the equation:

$$e_t = \rho e_{t-1} + v_t \quad (30)$$

- Assume the v_t are uncorrelated random errors with zero mean and constant variances:

$$E(v_t), \quad Var(v_t) = \sigma_v^2, \quad Cov(v_t, v_s) = 0 \text{ for } t \neq s \quad (31)$$

- Eq. 30 describes a *first-order autoregressive model* or a first-order autoregressive process for e_t .
- The term *AR(1) model* is used as an abbreviation for first-order autoregressive model.
- It is called an autoregressive model because it can be viewed as a regression model.
- It is called first-order because the right-hand-side variable is e_t lagged one period.
- We assume that:

$$-1 < \rho < 1 \quad (32)$$

- The mean and variance of e_t are:

$$E(e_t) = 0, \quad Var(e_t) = \sigma_e^2 = \frac{\sigma_v^2}{1 - \rho^2} \quad (33)$$

- The covariance term is:

$$Cov(e_t, e_{t-k}) = \frac{\rho^k \sigma_v^2}{1 - \rho^2}, \quad k > 0 \quad (34)$$

- The correlation implied by the covariance is:

$$\begin{aligned} \rho_k &= Corr(e_t, e_{t-k}) \\ &= \frac{Cov(e_t, e_{t-k})}{\sqrt{Var(e_t) Var(e_{t-k})}} \\ &= \frac{Cov(e_t, e_{t-k})}{Var(e_t)} \\ &= \frac{\rho^k \sigma_v^2 / (1 - \rho^2)}{\sigma_v^2 / (1 - \rho^2)} \\ &= \rho^k \end{aligned} \quad (35)$$

- Setting $k = 1$:

$$\rho_1 = \text{Cov}(e_t, e_{t-1}) = \rho \quad (36)$$

- ρ represents the correlation between two errors that are one period apart.
- It is the first-order autocorrelation for e , sometimes simply called the autocorrelation coefficient.
- It is the population autocorrelation at lag one for a time series that can be described by an $AR(1)$ model.
- r_1 is an estimate for ρ when we assume a series is $AR(1)$.
- Each e_t depends on all past values of the errors v_t :

$$e_t = v_t + \rho v_{t-1} + \rho^2 v_{t-2} + \rho^3 v_{t-3} + \dots$$

Example: Phillips Curve model (continued)

- For the Phillips Curve, we find for the first five lags:

$$\begin{aligned} r_1 &= 0.549, & r_2 &= 0.456, & r_3 &= 0.433, \\ r_4 &= 0.420, & r_5 &= 0.339 \end{aligned}$$

- For an $AR(1)$ model, we have:

$$\hat{\rho}_1 = \hat{\rho} = r_1 = 0.549$$

- For longer lags, we have:

$$\begin{aligned} \hat{\rho}_2 &= \hat{\rho}^2 = (0.549)^2 = 0.301 \\ \hat{\rho}_3 &= \hat{\rho}^3 = (0.549)^3 = 0.165 \\ \hat{\rho}_4 &= \hat{\rho}^4 = (0.549)^4 = 0.091 \\ \hat{\rho}_5 &= \hat{\rho}^5 = (0.549)^5 = 0.050 \end{aligned}$$

4.3 Nonlinear Least Squares Estimation

- Our model with an $AR(1)$ error is:

$$\begin{aligned} y_t &= \beta_1 + \beta_2 x_t + e_t \\ e_t &= \rho e_{t-1} + v_t \end{aligned} \quad (37)$$

with $-1 < \rho < 1$

- For v_t , we have:

$$E(v_t) = 0, \quad \text{Var}(v_t) = \sigma_v^2, \quad \text{Cov}(v_t, v_{t-1}) = 0 \text{ for } t \neq s \quad (38)$$

- With the appropriate substitutions, we get:

$$y_t = \beta_1 + \beta_2 x_t + \rho e_{t-1} + v_t \quad (39)$$

- For the previous period, the error is:

$$e_{t-1} = y_{t-1} - \beta_1 - \beta_2 x_{t-1} \quad (40)$$

- Multiplying by ρ :

$$\rho e_{t-1} = \rho y_{t-1} - \rho \beta_1 - \rho \beta_2 x_{t-1} \quad (41)$$

- Substituting, we get:

$$y_t = \beta_1 (1 - \rho) + \beta_2 x_t - \rho \beta_2 x_{t-1} + \rho y_{t-1} + v_t \quad (42)$$

- The coefficient of x_{t-1} equals $-\rho \beta_2$
- Although Eq. 42 is a linear function of the variables x_t , y_{t-1} and x_{t-1} , it is not a linear function of the parameters (β_1, β_2, ρ) .
- The usual linear least squares formulas cannot be obtained by using calculus to find the values of (β_1, β_2, ρ) that minimize S_v .
- These are nonlinear least squares estimates.

Example: Phillips Curve model (continued)

- Our Phillips Curve model assuming $AR(1)$ errors is:

$$INF_t = \beta_1 (1 - \rho) + \beta_2 DU_t - \rho \beta_2 DU_{t-1} + \rho INF_{t-1} + v_t \quad (43)$$

- Applying nonlinear least squares and presenting the estimates in terms of the original untransformed model, we have:

$$\begin{aligned} \widehat{INF} &= \begin{matrix} 0.7609 & -0.6944DU \\ (se) & (0.1245) & (0.3479) \end{matrix} \\ e_t &= \begin{matrix} 0.557e_{t-1} & +v_t \\ (0.090) \end{matrix} \end{aligned} \quad (44)$$

- Nonlinear least squares estimation of Eq. 43 is equivalent to using an iterative generalized least squares estimator called the Cochrane-Orcutt procedure.

4.4 Estimating a General Model

- We have the model:

$$y_t = \beta_1 (1 - \rho) + \beta_2 x_t - \rho \beta_2 x_{t-1} + \rho y_{t-1} + v_t \quad (45)$$

- Suppose now that we consider the model:

$$y_t = \delta + \theta_1 y_{t-1} + \delta_0 x_t + \delta_1 x_{t-1} + \rho + v_t \quad (46)$$

- This new notation will be convenient when we discuss a general class of autoregressive distributed lag (ARDL) models. Eq. 46 is a member of this class.
- Note that Eq. 46 is the same as Eq. 45 since:

$$\delta = \beta_1 (1 - \rho), \quad \theta_1 = \rho, \quad \delta_0 = \beta_2, \quad \delta_1 = -\rho \beta_2 \quad (47)$$

- Eq. 45 is a restricted version of Eq. 46 with the restriction from Eqs. 47 imposed.
- Applying the least squares estimator to Eq. 46 using the data for the Phillips curve example, yields:

$$\begin{aligned} \widehat{INF}_t &= \begin{matrix} 0.3336 & +0.5593INF_{t-1} \\ (se) & (0.0899) & (0.0908) \end{matrix} \\ &\quad \begin{matrix} -0.6882DU_t & +0.320DU_{t-1} \\ (0.2575) & (0.2499) \end{matrix} \end{aligned} \quad (48)$$

- The equivalent AR(1) estimates are:

$$\begin{aligned}\hat{\delta} &= \hat{\beta}_1 (1 - \hat{\rho}) = 0.7609 \times (1 - 0.5574) = 0.3368 \\ \hat{\theta}_1 &= \hat{\rho} = 0.5574 \\ \hat{\delta}_0 &= \hat{\beta}_2 = -0.6944 \\ \hat{\delta}_1 &= -\hat{\rho}\hat{\beta}_2 = -0.5574 \times (-0.6944) = 0.3871\end{aligned}$$

- These are similar to our other estimates.
- The original economic model for the Phillips Curve was:

$$INF_t = INF_t^E - \gamma (U_t - U_{t-1}) \quad (49)$$

- Re-estimation of the model after omitting DU_{t-1} yields:

$$\begin{array}{rcccl}\widehat{INF}_t = & 0.3548 & +0.5282INF_{t-1} & -0.4909DU_t & \\ (se) & (0.0876) & (0.0851) & (0.1921) & \end{array} \quad (50)$$

- In this model, inflationary expectations are given by:

$$\widehat{INF}_t^E = 0.3548 + 0.5282INF_{t-1}$$

- A 1% rise in the unemployment rate is expected to lead to an approximate 0.5% fall in the inflation rate.
- We have described three ways of overcoming the effect of serially correlated errors:
 1. Estimate the model using least squares with *HAC* standard errors.
 2. Use nonlinear least squares to estimate the model with a lagged x , a lagged y , and the restriction implied by an *AR*(1) error specification.
 3. Use least squares to estimate the model with a lagged x and a lagged y , but without the restriction implied by an *AR*(1) error specification.

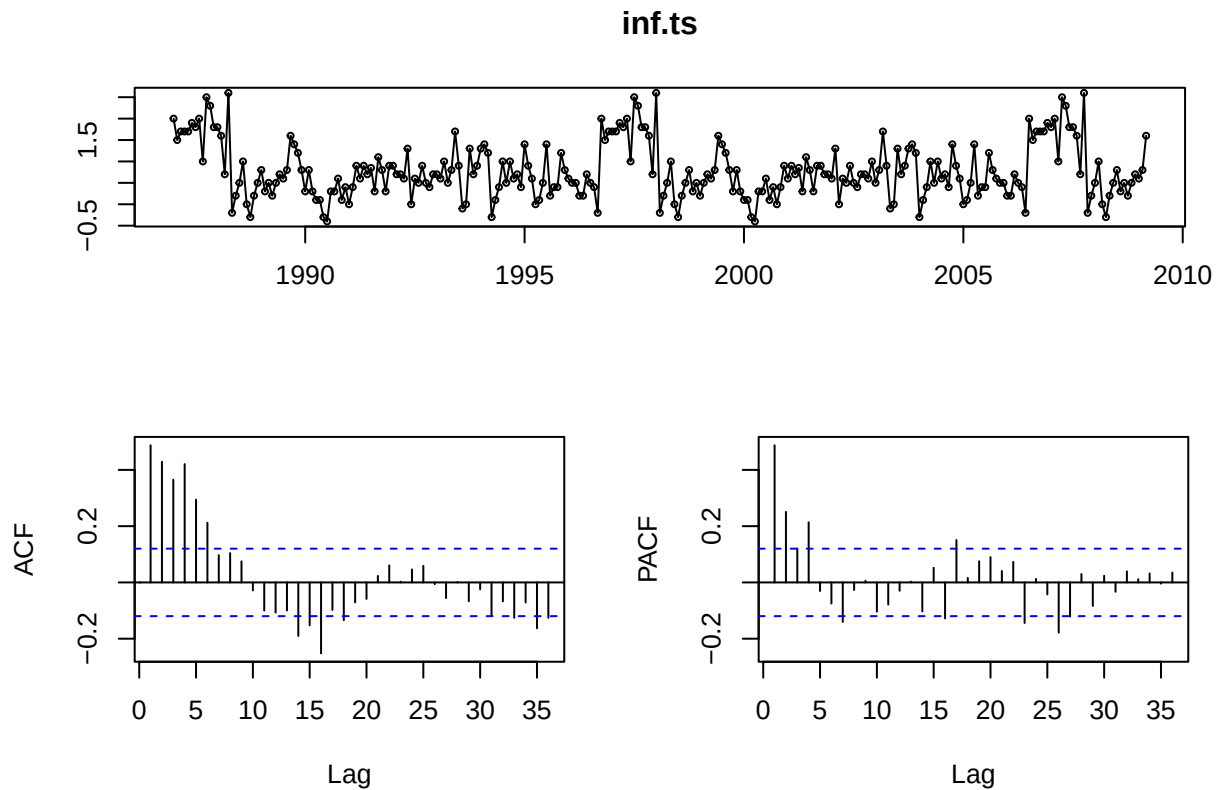
R Example

```
# phillips: monthly (1987:01 - 2009:3) observations on
#   inf = Australian Inflation Rate (Percentage Change in the Consumer Price Index
#         with an adjustment for 2000Q3 when a national sales tax was introduced)
#   u = Australian Unemployment Rate (Seasonally adjusted)
library(POE5Rdata)
library(tseries)
library(forecast)
library(lmtest)
data("phillips5_aus")
attach(phillips5_aus)
library(dynlm)
```



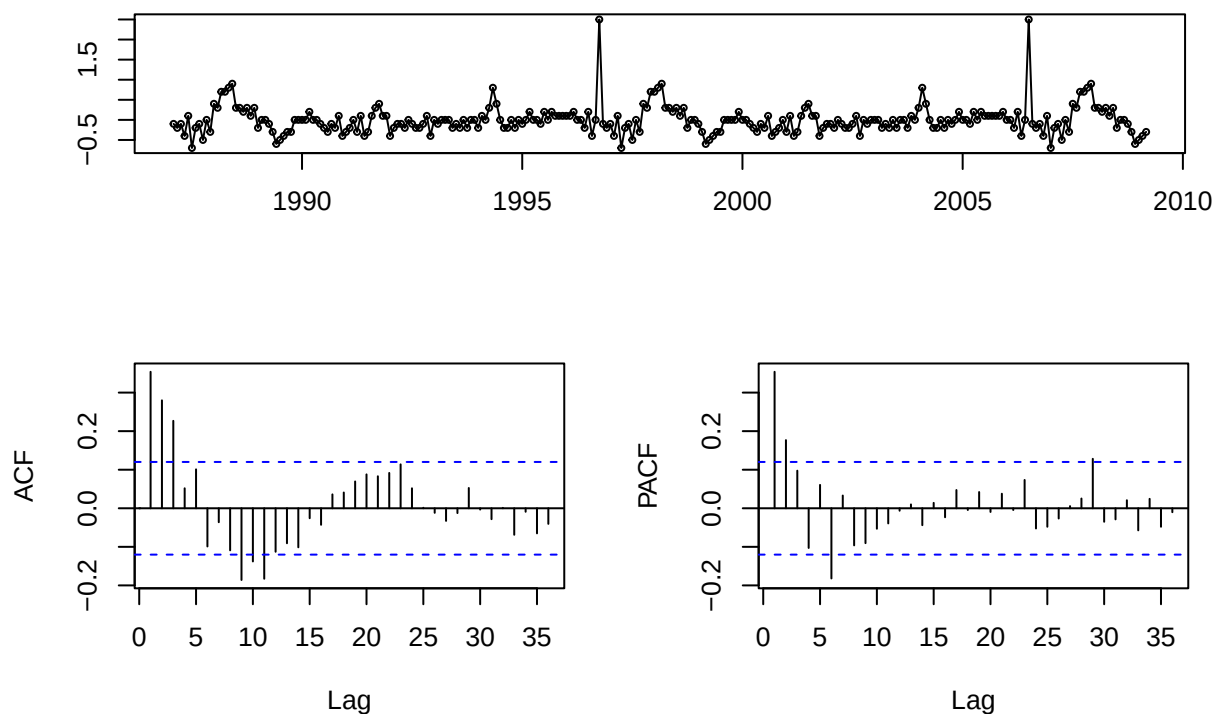
```
# (1) Compare Incorrect vs. HAC std. Errors
```

```
phillips_au.ts <- ts(phillips5_au,  
  start=c(1987,1),  
  end=c(2009,3),  
  frequency=12)  
inf.ts <- phillips_au.ts[, "inf"]  
Du.ts <- diff(phillips_au.ts[, "u"])  
  
tsdisplay(inf.ts)
```



```
tsdisplay(Du.ts)
```

Du.ts



```
# Regress inflation of unemployment rate
reg.ts = dynlm(inf.ts~Du.ts,data=phillips5_au.ts)
```

```
library(sandwich)
coeftest(reg.ts) # Incorrect se
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.781337   0.039597 19.7322  <2e-16 ***
## Du.ts        -0.088930   0.109162  -0.8147   0.416
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(reg.ts, vcov.=NeweyWest(reg.ts)) # Newey-West se
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.781337   0.069398 11.2587  <2e-16 ***
## Du.ts        -0.088930   0.180253  -0.4934   0.6222
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# (2) Non-Linear Least Squares: Cochrane-Orcutt
```

```
data.ts.table <- cbind(phillips5_au.ts[, "inf"],
                      phillips5_au.ts[, "u"],
```

```

stats::lag(phillips5_aus.ts[, "inf"], -1),
diff(phillips5_aus.ts[, "u"], lag=1),
stats::lag(diff(phillips5_aus.ts[, 2], lag=1), -1))
dfr <- data.frame(data.ts.table)
names(dfr) <- c("inf", "u", "Linf", "Du", "LDu")
reg.ts.nls <- nls(inf~b1*(1-rho)+b2*Du+rho*Linf-rho*b2*LDu,
  data=dfr,
  start=list(rho=0.5, b1=0.5, b2=-0.5))
summary(reg.ts.nls)

```

```

##
## Formula: inf ~ b1 * (1 - rho) + b2 * Du + rho * Linf - rho * b2 * LDu
##
## Parameters:
##      Estimate Std. Error t value Pr(>|t|)
## rho  0.48787    0.05417   9.006  <2e-16 ***
## b1   0.77909    0.06772  11.504  <2e-16 ***
## b2  -0.05043    0.09264  -0.544    0.587
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5646 on 262 degrees of freedom
##
## Number of iterations to convergence: 4
## Achieved convergence tolerance: 2.802e-06
## (3 observations deleted due to missingness)

```

```
summary(reg.ts)
```

```

##
## Time series regression with "ts" data:
## Start = 1987(2), End = 2009(3)
##
## Call:
## dynlm(formula = inf.ts ~ Du.ts, data = phillips5_aus.ts)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.1991 -0.4519 -0.1258  0.3453  1.8809
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.78134    0.03960  19.732  <2e-16 ***
## Du.ts       -0.08893    0.10916  -0.815    0.416
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6458 on 264 degrees of freedom
## Multiple R-squared:  0.002508, Adjusted R-squared: -0.001271
## F-statistic: 0.6637 on 1 and 264 DF, p-value: 0.416

```

5 Autoregressive Distributed Lag (ARDL) Models

- An autoregressive distributed lag ($ARDL(p, q)$) model is one that contains both lagged x_t 's and lagged y_t 's. The lag order of y_t is p , and for x_t it is q .

$$y_t = \delta + \delta_0 x_t + \delta_1 x_{t-1} + \cdots + \delta_q x_{t-q} + \theta_1 y_{t-1} + \cdots + \theta_p y_{t-p} + v_t \quad (51)$$

- Two examples:

$ARDL(1, 1)$:

$$\widehat{INF}_t = 0.3336 + 0.5593 INF_{t-1} - 0.6882 DU_t + 0.320 DU_{t-1}$$

$ARDL(1, 0)$:

$$\widehat{INF}_t = 0.3548 + 0.5282 INF_{t-1} - 0.4909 DU_t$$

- An $ARDL$ model can be transformed into one with only lagged x 's which go back into the infinite past:

$$\begin{aligned} y_t &= \alpha + \delta_0 x_t + \beta_1 x_{t-1} + \beta_2 x_{t-2} + \cdots + e_t \\ &= \alpha + \sum_{s=0}^{\infty} \beta_s x_{t-s} + e_t \end{aligned} \quad (52)$$

- This model is called an *infinite distributed lag model*.
- Four possible criteria for choosing p and q :
 - Has serial correlation in the errors been eliminated?
 - Are the signs and magnitudes of the estimates consistent with our expectations from economic theory?
 - Are the estimates significantly different from zero, particularly those at the longest lags?
 - What values for p and q minimize information criteria such as the AIC and SC ?
- The *Akaike information criterion* (AIC) is:

$$AIC = \ln \left(\frac{SSE}{T} \right) + \frac{2K}{T} \quad (53)$$

where $K = p + q + 2$.

- The *Schwarz criterion* (SC), also known as the Bayes information criterion (BIC), is:

$$SC = \ln \left(\frac{SSE}{T} \right) + \frac{K \ln(T)}{T} \quad (54)$$

- Because $K \ln(T) / T > 2K/T$ for $T \geq 8$, the SC penalizes additional lags more heavily than does the AIC .

Example: Phillips Curve model (continued)

- Consider the previously estimated $ARDL(1, 0)$ model:

$$\begin{array}{ccccccc} \widehat{INF}_t = & 0.3548 & +0.5282 INF_{t-1} & -0.4909 DU_t & , & Obs = 90 \\ (se) & (0.0876) & (0.0851) & (0.1921) & & \end{array} \quad (55)$$

- For an $ARDL(4, 0)$ version of the model ($Obs = 87$):

$$\begin{array}{ccccccc} \widehat{INF}_t = & 0.1001 & +0.2354 INF_{t-1} & +0.1213 INF_{t-2} & & & \\ (se) & (0.0983) & (0.1016) & (0.1038) & & & \\ & +0.1677 INF_{t-3} & +0.2819 INF_{t-4} & -0.7902 DU_t & & & \\ & (0.1050) & (0.1014) & (0.1885) & & & \end{array} \quad (56)$$

- Inflationary expectations are given by:

$$\widehat{INF}_t^E = 0.1001 + 0.2354INF_{t-1} + 0.1213INF_{t-2} + 0.1677INF_{t-3} + 0.2819INF_{t-4}$$

Example: Recall the model for Okun's Law,

$$\widehat{DU}_t = 0.5836 - 0.2020G_t - 0.1653G_{t-1} - 0.070G_{t-2}, \quad Obs = 96 \quad (57)$$

- Now consider this version:

$$\widehat{DU}_t = \begin{matrix} 0.3780 & +0.3501DU_{t-1} & -0.1841G_t \\ (se) & (0.0578) & (0.0846) & (0.0307) \end{matrix} \quad (58)$$

$$\begin{matrix} -0.0992G_{t-1} \\ (0.0368) \end{matrix}, \quad Obs = 96 \quad (59)$$

Which model is better? Why?

5.1 Autoregressive Models: AR(p)

- An autoregressive model of order p , denoted $AR(p)$, is given by:

$$y_t = \delta + \theta_1 y_{t-1} + \theta_2 y_{t-2} + \cdots + \theta_p y_{t-p} + v_t \quad (60)$$

Example: Consider a model for growth in real GDP,

$$\widehat{G}_t = \begin{matrix} 0.4657 & +0.377G_{t-1} & +0.2462G_{t-2} \\ (se) & (0.1433) & (0.1000) & (0.1029) \end{matrix}, \quad Obs = 96 \quad (61)$$

This model is an AR(2), and based on the estimates, we should keep the two lags of G_t . We can estimate different AR(p) models, and use AIC and/or BIC to identify the best fit model.

R Example

```
# Okun: quarterly (1985Q2 - 2009Q3) observations
# g = percentage change in U.S. Gross Domestic Product, seasonally adjusted
# u = U.S. Civilian Unemployment Rate (Seasonally adjusted)
library(POE5Rdata)
library(tseries)
library(forecast)
data("okun5_aus")
attach(okun5_aus)

# Convert to time series format
g_ts = ts(g,start=c(1985,2), end=c(2009,3), frequency=4)

# Regress DU_t on present and past growth rates G_{t}, G_{t-1}, ..., G_{t-5}
library(dynlm)
m1 = dynlm(g_ts~L(g_ts,1))
m2 = dynlm(g_ts~L(g_ts,1:2))
m3 = dynlm(g_ts~L(g_ts,1:3))
m4 = dynlm(g_ts~L(g_ts,1:4))
m5 = dynlm(g_ts~L(g_ts,1:5))
#summary(reg.ts)
AIC(m1, m2, m3, m4, m5)
```

```
##      df      AIC
## m1   3 262.3486
## m2   4 261.5895
## m3   5 261.9048
## m4   6 256.5689
## m5   7 247.1855
```

Model 4 seems like the best model based on AIC. However, we have to look at other diagnostics.

6 Forecasting

- We consider forecasting using three different models:
 1. AR model
 2. ARDL model
 3. Exponential smoothing model

6.1 Forecasting with AR Models

Example: Consider again the model for growth in real GDP,

- Consider an AR(2) model for real GDP growth:

$$G_t = \delta + \theta_1 G_{t-1} + \theta_2 G_{t-2} + v_t \quad (62)$$

- The model to forecast G_{T+1} is:

$$G_{T+1} = \delta + \theta_1 G_T + \theta_2 G_{T-1} + v_{T+1}$$

- The growth values for the two most recent quarters are:

$$\begin{aligned} G_T &= G_{2009Q3} = 0.8 \\ G_{T-1} &= G_{2009Q2} = -0.2 \end{aligned}$$

- The forecast for G_{2009Q4} is:

$$\begin{aligned} \hat{G}_{T+1} &= \hat{\delta} + \hat{\theta}_1 G_T + \hat{\theta}_2 G_{T-1} \\ &= 0.46573 + 0.3770 \times 0.8 + 0.24624 \times (-0.2) \\ &= 0.7181 \end{aligned} \quad (63)$$

- For two quarters ahead, the forecast for G_{2010Q1} is:

$$\begin{aligned} \hat{G}_{T+2} &= \hat{\delta} + \hat{\theta}_1 G_{T+1} + \hat{\theta}_2 G_T \\ &= 0.46573 + 0.3770 \times 0.7181 + 0.24624 \times 0.8 \\ &= 0.9334 \end{aligned} \quad (64)$$

- For three periods out, it is:

$$\begin{aligned} \hat{G}_{T+3} &= \hat{\delta} + \hat{\theta}_1 G_{T+2} + \hat{\theta}_2 G_{T+1} \\ &= 0.46573 + 0.3770 \times 0.9334 + 0.24624 \times 0.7181 \\ &= 0.9945 \end{aligned} \quad (65)$$

- Summarizing our forecasts:

Real GDP growth rates for 2009Q4, 2010Q1, and 2010Q2 are approximately 0.72%, 0.93%, and 0.99%, respectively.

- A 95% interval forecast for j periods into the future is given by:

$$\hat{G}_{T+j} \pm t_{(0.975, df)} \hat{\sigma}_j$$

where $\hat{\sigma}_j$ is the standard error of the forecast error and df is the number of degrees of freedom in the estimation of the AR model.

- The first forecast error, occurring at time $T + 1$, is:

$$\begin{aligned} u_1 &= G_{T+1} - \hat{G}_{T+1} = (\delta - \hat{\delta}) \\ &\quad + (\theta_1 - \hat{\theta}_1) G_T + (\theta_2 - \hat{\theta}_2) G_{T-1} + v_{T+1} \end{aligned}$$

- Ignoring the error from estimating the coefficients, we get:

$$u_1 = v_{T+1} \tag{66}$$

- The forecast error for two periods ahead is:

$$u_2 = \theta_1 (G_{T+1} - \hat{G}_{T+1}) + v_{T+2} = \theta_1 u_1 + v_{T+2} = \theta_1 v_{T+1} + v_{T+2} \tag{67}$$

- The forecast error for three periods ahead is:

$$u_3 = \theta_1 u_1 + \theta_2 u_2 + v_{T+3} = (\theta_1^2 + \theta_2) v_{T+1} + \theta_1 v_{T+2} + v_{T+3} \tag{68}$$

- Because the v_t 's are uncorrelated with constant variance σ_v^2 , we can show that:

$$\begin{aligned} \sigma_1^2 &= \text{Var}(u_1) = \sigma_v^2 \\ \sigma_2^2 &= \text{Var}(u_2) = \sigma_v^2 (1 + \theta_1^2) \\ \sigma_3^2 &= \text{Var}(u_3) = \sigma_v^2 \left[(\theta_1^2 + \theta_2)^2 + \theta_1^2 + 1 \right] \end{aligned}$$

Quarter	Forecast $\hat{G}_{T+j}$	Standard Error of Forecast Error ($\hat{\sigma}_j$)	Forecast Interval ($\hat{G}_{T+j} \pm 1.9858 \times \hat{\sigma}_j$)
2009Q4 ($j = 1$)	0.71808	0.55269	(−0.379, 1.816)
2010Q1 ($j = 2$)	0.93343	0.59066	(−0.239, 2.106)
2010Q2 ($j = 3$)	0.99445	0.62845	(−0.254, 2.242)

Figure 8: Forecasts and Forecast Intervals for GDP Growth

6.2 Forecasting with ARDL Models

- Consider forecasting future unemployment using Okun's Law with an $ARDL(1, 1)$:

$$DU_t = \delta + \theta_1 DU_{t-1} + \delta_0 G_t + \delta_1 G_{t-1} + v_t \quad (69)$$

- The value of DU in the first post-sample quarter is:

$$DU_{T+1} = \delta + \theta_1 DU_T + \delta_0 G_{T+1} + \delta_1 G_T + v_{T+1} \quad (70)$$

- But we need a value for G_{T+1}
- Now consider the *change* in unemployment
- Rewrite Eq. 70 as:

$$U_{T+1} - U_T = \delta + \theta_1 (U_T - U_{T-1}) + \delta_0 G_{T+1} + \delta_1 G_T + v_{T+1}$$

- Rearranging:

$$U_{T+1} = \delta + (\theta_1 + 1) U_T - \theta_1 U_{T-1} + \delta_0 G_{T+1} + \delta_1 G_T + v_{T+1} \quad (71)$$

$$= \delta + \theta_1^* U_T - \theta_2^* U_{T-1} + \delta_0 G_{T+1} + \delta_1 G_T + v_{T+1} \quad (72)$$

- For the purpose of computing point and interval forecasts, the $ARDL(1, 1)$ model for a change in unemployment can be written as an $ARDL(2, 1)$ model for the level of unemployment.
- This result holds not only for $ARDL$ models where a dependent variable is measured in terms of a change or difference, but also for pure AR models involving such variables.

6.3 Forecasting with Exponential Smoothing Models

- Another popular model used for predicting the future value of a variable on the basis of its history is the exponential smoothing method.
- Like forecasting with an AR model, forecasting using exponential smoothing does not utilize information from any other variable.
- One possible forecasting method is to use the average of past information, such as:

$$\hat{y}_{T+1} = \frac{y_T + y_{T-1} + y_{T-2}}{3}$$

- This forecasting rule is an example of a simple (equally-weighted) moving average model with $k = 3$, $MA(3)$.
- Now consider a form in which the weights decline exponentially as the observations get older:

$$\hat{y}_{T+1} = \alpha y_T + \alpha (1 - \alpha)^1 y_{T-1} + \alpha (1 - \alpha)^2 y_{T-2} + \dots \quad (73)$$

- We assume that $0 < \alpha < 1$.
- Also, it can be shown that: $\sum_{s=0}^{\infty} \alpha (1 - \alpha)^s = 1$.
- For forecasting, recognize that:

$$(1 - \alpha) \hat{y}_T = \alpha (1 - \alpha) y_{T-1} + \alpha (1 - \alpha)^2 y_{T-2} + \alpha (1 - \alpha)^3 y_{T-3} + \dots \quad (74)$$

- We can simplify Eq. 74 to:

$$\hat{y}_{T+1} = \alpha y_T + (1 - \alpha) \hat{y}_T \quad (75)$$

- The value of α can reflect one's judgment about the relative weight of current information.
- It can be estimated from historical information by obtaining *within-sample forecasts*:

$$\hat{y}_t = \alpha y_{t-1} + (1 - \alpha) \hat{y}_{t-1}, \quad t = 2, 3, \dots, T \quad (76)$$

- Choosing α that minimizes the sum of squares of the *one-step forecast errors*:

$$v_t = y_t - (\alpha y_{t-1} + (1 - \alpha) \hat{y}_{t-1}) \quad (77)$$

Example: Consider again the model for growth in real GDP,

- The forecasts for 2009Q4 from each value of α are:

$$\begin{aligned} \alpha = 0.38: \hat{G}_{T+1} &= \alpha G_T + (1 - \alpha) \hat{G}_T \\ &= 0.38 \times 0.8 + (1 - 0.38) \times (-0.40392) \\ &= 0.0536 \\ \alpha = 0.38: \hat{G}_{T+1} &= \alpha G_T + (1 - \alpha) \hat{G}_T \\ &= 0.8 \times 0.8 + (1 - 0.8) \times (-0.392358) \\ &= 0.5613 \end{aligned}$$

R Example

```
# Forecasting Examples using (1) AR and (2) Exp. Smoothing Models (ETS, and Holt-Winters)
library(PoEdata)
library(tseries)
library(forecast)
data("okun5_aus")
attach(okun5_aus)
library(dynlm)

# 1. Forecast from AR(2)
okun.ts <- ts(okun5_aus, start=c(1985,2), end=c(2009,3), frequency=4)
y <- okun.ts[, "g"]

# (a) Estimate the AR(2) Model
reg.ar2 = ar(y, aic=FALSE, order.max=2, method="ols")
summary(reg.ar2)
```

##	Length	Class	Mode
## order	1	-none-	numeric
## ar	2	-none-	numeric
## var.pred	1	-none-	numeric
## x.mean	1	-none-	numeric
## x.intercept	1	-none-	numeric

```
## aic          1      -none- numeric
## n.used       1      -none- numeric
## n.obs        1      -none- numeric
## order.max    1      -none- numeric
## partialacf   0      -none- NULL
## resid        98     ts      numeric
## method       1      -none- character
## series       1      -none- character
## frequency    1      -none- numeric
## call         5      -none- call
## asy.se.coef  2      -none- list
```

```
print(reg.ar2)
```

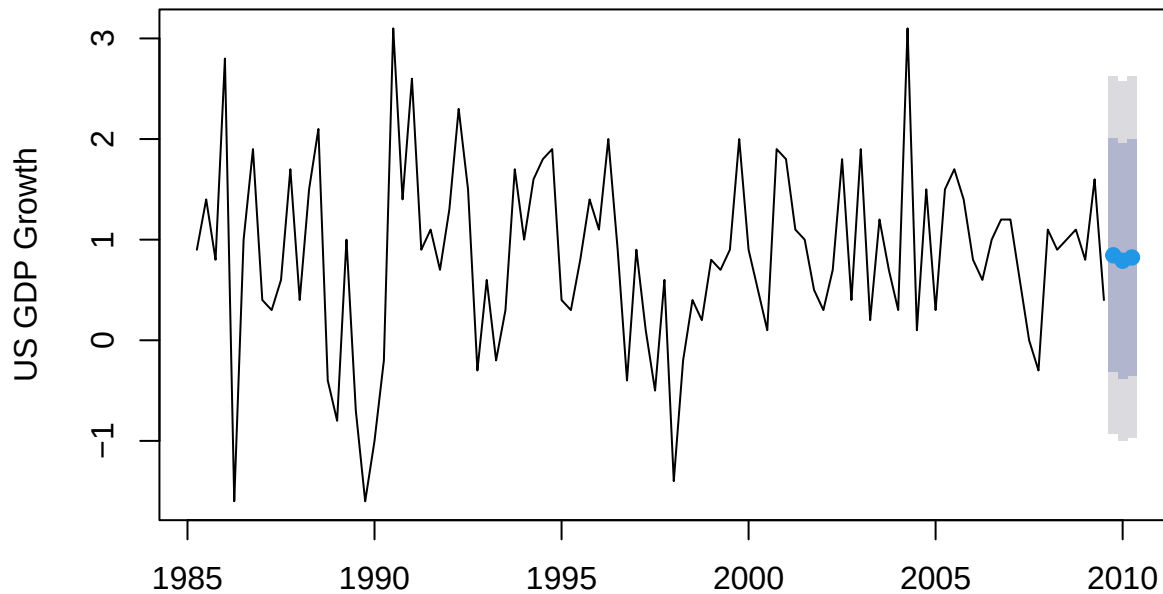
```
##
## Call:
## ar(x = y, aic = FALSE, order.max = 2, method = "ols")
##
## Coefficients:
##      1      2
## 0.1135 0.0873
##
## Intercept: -0.006722 (0.09252)
##
## Order selected 2  sigma^2 estimated as  0.8217
```

```
# (b) Forecast 3 steps-ahead (include point estimates and respective 95% CI )
forecast(reg.ar2, 3)
```

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 2009 Q4	0.8446769	-0.3170544	2.006408	-0.9320378	2.621392
## 2010 Q1	0.7903195	-0.3788679	1.959507	-0.9977984	2.578437
## 2010 Q2	0.8229932	-0.3519777	1.997964	-0.9739698	2.619956

```
# (c) Plot the original series with the respective forecast
plot(forecast(reg.ar2, 3), ylab = "US GDP Growth")
```

Forecasts from AR(2)



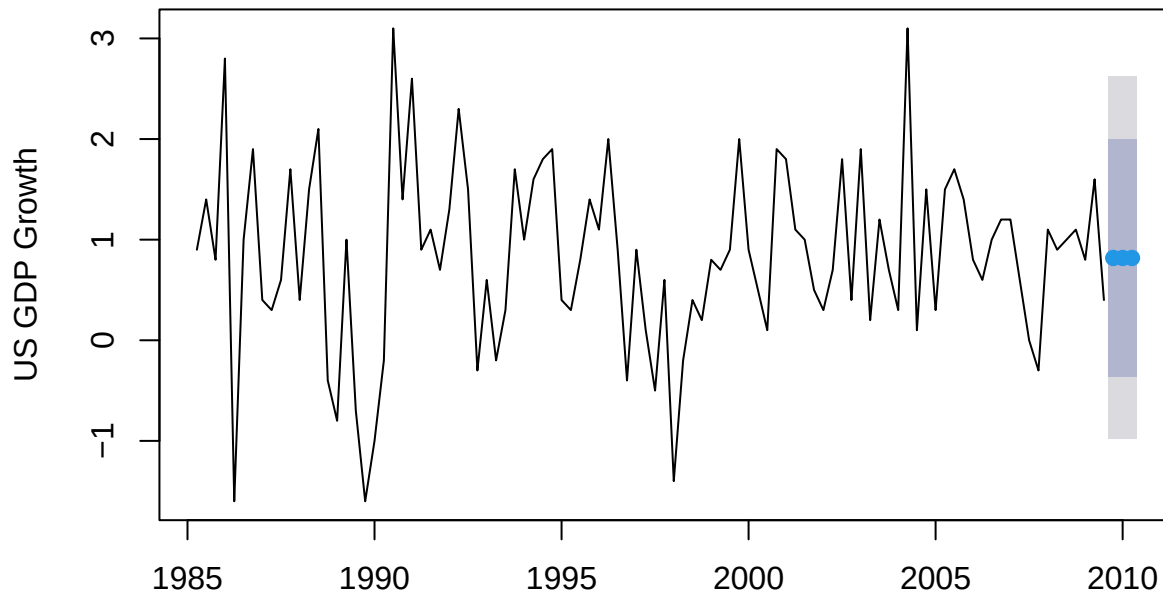
```
# 2. Forecast from Exp. Smoothing (ETS)
# Here are two great online resources:
# https://otexts.com/fpp2/
# https://www.econometrics-with-r.org/index.html
```

```
forecast(y,h=3)
```

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 2009 Q4	0.8182184	-0.3597655	1.996202	-0.9833525	2.619789
## 2010 Q1	0.8182184	-0.3597655	1.996202	-0.9833526	2.619789
## 2010 Q2	0.8182184	-0.3597655	1.996202	-0.9833526	2.619789

```
plot(forecast(y,h=3),ylab = "US GDP Growth")
```

Forecasts from ETS(A,N,N)



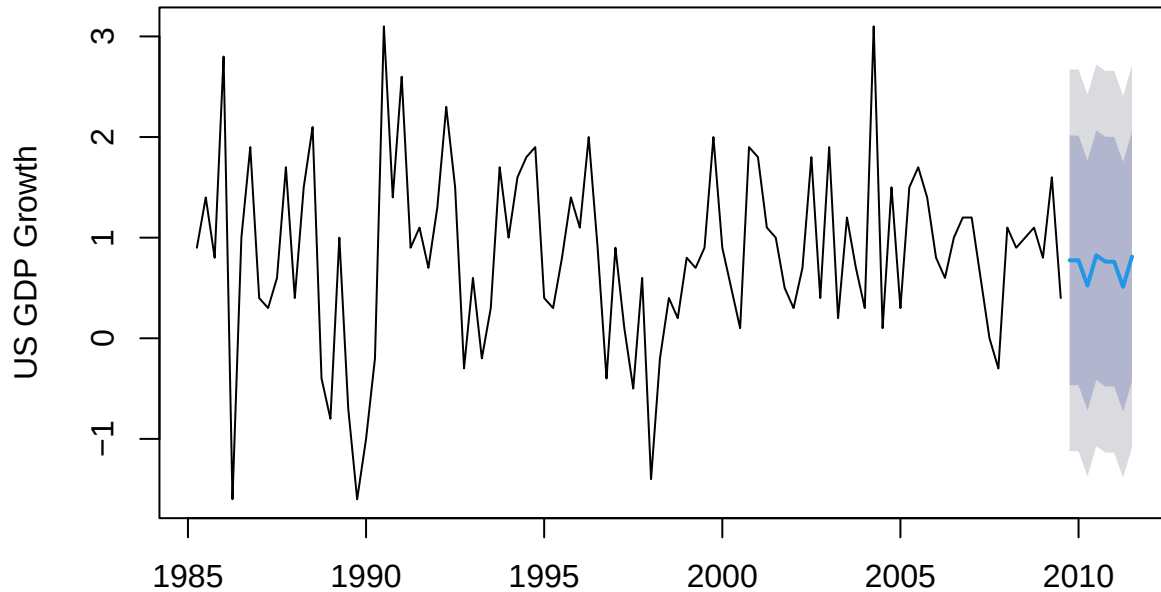
```
# 2. Forecast from Holt-Winters
```

```
hw(y,seasonal="additive")
```

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 2009 Q4	0.7755405	-0.4648994	2.015980	-1.121549	2.672630
## 2010 Q1	0.7737957	-0.4666443	2.014236	-1.123294	2.670885
## 2010 Q2	0.5248121	-0.7156280	1.765252	-1.372277	2.421901
## 2010 Q3	0.8246897	-0.4157505	2.065130	-1.072400	2.721779
## 2010 Q4	0.7623133	-0.4781273	2.002754	-1.134777	2.659404
## 2011 Q1	0.7605685	-0.4798724	2.001009	-1.136522	2.657659
## 2011 Q2	0.5115849	-0.7288565	1.752026	-1.385506	2.408676
## 2011 Q3	0.8114625	-0.4289794	2.051904	-1.085630	2.708555

```
plot(hw(y,seasonal="additive"),ylab = "US GDP Growth")
```

Forecasts from Holt–Winters' additive method



7 Multiplier Analysis

This section will not be covered in the course. However, it is included for completeness in case you are interested in learning more about the topic.

- Multiplier analysis refers to the effect, and the timing of the effect, of a change in one variable on the outcome of another variable.
- Let's find multipliers for an *ARDL* model of the form:

$$y_t = \delta + \theta_1 y_{t-1} + \cdots + \theta_p y_{t-p} + \delta_0 x_t + \delta_1 x_{t-1} + \cdots + \delta_q x_{t-q} + v_t \quad (78)$$

- We can transform this into an infinite distributed lag model:

$$y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \beta_2 x_{t-2} + \cdots + v_t \quad (79)$$

- The multipliers are defined as:

$$\beta_s = \frac{\partial y_t}{\partial x_{t-s}} = s \text{ period delay multiplier}$$

$$\sum_{j=0}^s \beta_j = s \text{ period interim multiplier}$$

$$\sum_{j=0}^{\infty} \beta_j = s \text{ period total multiplier}$$

- The lag operator is defined as:

$$Ly_t = y_{t-1}$$

- Lagging twice, we have:

$$L(Ly_t) = Ly_{t-1} = y_{t-2}$$

Or:

$$L^2y_t = y_{t-2}$$

- More generally, we have:

$$L^s y_t = y_{t-s}$$

- Now, rewrite our model as:

$$y_t = \delta + \theta_1 Ly_t + \cdots + \theta_p L^p y_t + \delta_0 x_t + \delta_1 Lx_t + \cdots + \delta_q L^q x_t + v_t \quad (80)$$

- Rearranging terms:

$$(1 - \theta_1 L - \cdots - \theta_p L^p) y_t = \delta + (\delta_0 + \delta_1 L + \cdots + \delta_q L^q) x_t + v_t \quad (81)$$

- Let's apply this to our Okun's Law model

- The model:

$$DU_t = \delta + \theta_1 DU_{t-1} + \delta_0 G_t + \delta_1 G_{t-1} + v_t \quad (82)$$

can be rewritten as:

$$(1 - \theta_1 L) DU_t = \delta + (\delta_0 + \delta_1 L) G_t + v_t \quad (83)$$

- Define the inverse of $(1 - \theta_1 L)$ as $(1 - \theta_1 L)^{-1}$ such that:

$$(1 - \theta_1 L)^{-1} (1 - \theta_1 L) = 1$$

- Multiply both sides of Eq. 83 by $(1 - \theta_1 L)^{-1}$:

$$DU_t = (1 - \theta_1 L)^{-1} \delta + (1 - \theta_1 L)^{-1} (\delta_0 + \delta_1 L) G_t + (1 - \theta_1 L)^{-1} v_t \quad (84)$$

- Equating this with the infinite distributed lag representation:

$$\begin{aligned} DU_t &= \alpha + \beta_0 G_t + \beta_1 G_{t-1} + \cdots + e_t \\ &= \alpha + (\beta_0 + \beta_1 L + \beta_2 L^2 + \cdots) G_t + e_t \end{aligned} \quad (85)$$

- For Eqs. 82 and 85 to be identical, it must be true that:

$$\alpha = (1 - \theta_1 L)^{-1} \delta \quad (86)$$

$$\beta_0 + \beta_1 L + \beta_2 L^2 + \cdots = (1 - \theta_1 L)^{-1} (\delta_0 + \delta_1 L) \quad (87)$$

$$e_t = (1 - \theta_1 L)^{-1} v_t \quad (88)$$

- Multiply both sides of Eq. 86 by $(1 - \theta_1 L)$ to obtain $(1 - \theta_1 L)\alpha = \delta$

- Note that the lag of a constant that does not change so $L\alpha = \alpha$

- Now we have:

$$\begin{aligned} (1 - \theta_1) \alpha &= \delta \quad \text{and} \\ \alpha &= \frac{\delta}{1 - \theta_1} \end{aligned}$$

- Multiplying both sides of Eq. 87 by $(1 - \theta_1 L)$:

$$\begin{aligned}
(\delta_0 + \delta_1 L) &= (1 - \theta_1 L) (\beta_0 + \beta_1 L + \beta_2 L^2 + \dots) \\
&= \beta_0 + \beta_1 L + \beta_2 L^2 + \dots \\
&\quad - \beta_0 \theta_1 L - \beta_1 \theta_1 L^2 - \beta_2 \theta_1 L^3 - \dots \\
&= \beta_0 + (\beta_1 - \beta_0 \theta_1) L + (\beta_2 - \beta_1 \theta_1) L^2 + (\beta_3 - \beta_2 \theta_1) L^3 + \dots
\end{aligned} \tag{89}$$

- Rewriting Eq. 89 as:

$$\delta_0 + \delta_1 L + 0L^2 + 0L^3 = \beta_0 + (\beta_1 - \beta_0 \theta_1) L + (\beta_2 - \beta_1 \theta_1) L^2 + (\beta_3 - \beta_2 \theta_1) L^3 + \dots \tag{90}$$

- Equating coefficients of like powers in L yields:

$$\begin{aligned}
\delta_0 &= \beta_0 \\
\delta_1 &= (\beta_1 - \beta_0 \theta_1) \\
0 &= (\beta_2 - \beta_1 \theta_1) \\
0 &= (\beta_3 - \beta_2 \theta_1) \dots
\end{aligned}$$

and so on.

- We can now find the β 's using the recursive equations:

$$\begin{aligned}
\beta_0 &= \delta_0 \\
\beta_1 &= \delta_1 + \beta_0 \theta_1 \\
\beta_j &= \beta_{j-1} \theta_1, \text{ for all } j \geq 2
\end{aligned} \tag{91}$$

- You can start from the equivalent of Eq. 90 which, in its general form, is:

$$\begin{aligned}
\delta_0 + \delta_1 L + \delta_2 L^2 + \dots + \delta_q L^q &= (1 - \theta_1 L - \theta_2 L^2 - \dots - \theta_p L^p) \\
&\quad \times (\beta_0 + \beta_1 L + \beta_2 L^2 + \dots)
\end{aligned} \tag{92}$$

- Given the values p and q for your *ARDL* model, you need to multiply out the above expression, and then equate coefficients of like powers in the lag operator.

Example: For Okun's Law model,

$$\widehat{DU}_t = 0.37 + 0.3501DU_{t-1} - 0.1841G_t - 0.0992G_{t-1}$$

- The impact and delay multipliers for the first four quarters are:

$$\begin{aligned}\widehat{\beta}_0 &= \widehat{\delta}_0 = -0.1841 \\ \widehat{\beta}_1 &= \widehat{\delta}_1 + \widehat{\beta}_0\widehat{\theta}_1 = -0.0992 - 0.1841 \times 0.3501 = -0.1636 \\ \widehat{\beta}_2 &= \widehat{\beta}_1\widehat{\theta}_1 = -0.1636 \times 0.3501 = -0.0573 \\ \widehat{\beta}_3 &= \widehat{\beta}_2\widehat{\theta}_1 = -0.0573 \times 0.3501 = -0.0201 \\ \widehat{\beta}_4 &= \widehat{\beta}_3\widehat{\theta}_1 = -0.0201 \times 0.3501 = -0.0070\end{aligned}$$

- We can estimate the total multiplier given by:

$$\sum_{j=0}^{\infty} \beta_j$$

and the normal growth rate that is needed to maintain a constant rate of unemployment:

$$G_N = -\frac{\alpha}{\sum_{j=0}^{\infty} \beta_j}$$

- We can show that:

$$\sum_{j=0}^{\infty} \beta_j = \widehat{\delta}_0 + \frac{\widehat{\delta}_1 + \widehat{\delta}_0\widehat{\theta}_1}{1 - \widehat{\theta}_1} = -0.1841 + \frac{-0.1636}{1 - 0.3501} = -0.4358$$

- An estimate for α is given by:

$$\widehat{\alpha} = \frac{\widehat{\delta}}{1 - \widehat{\theta}_1} = \frac{0.37801}{1 - 0.649884} = 0.5817$$

- Therefore, normal growth rate is:

$$\widehat{G}_N = \frac{0.5817}{0.4358} = 1.3\% \text{ per quarter}$$

As seen in the figure below, an increase in GDP growth leads to a fall in unemployment, with its greatest effect being felt in the current and next quarters and a declining effect thereafter.

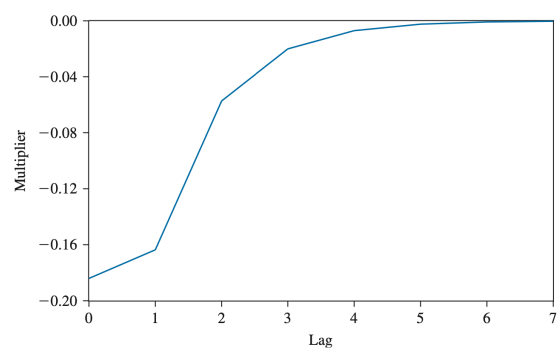


Figure 9: Delay multipliers from Okun's law ARDL(1,1) model.

R Example

```
# Okun's ARDL and Multipliers Example
library(PoEdata)
library(tseries)
library(forecast)
library(dynlm)
data("okun5_aus")
attach(okun5_aus)

# Convert to time series format
g_ts = ts(g,start=c(1985,2), end=c(2009,3),frequency=4)
u_ts = ts(u,start=c(1985,2), end=c(2009,3),frequency=4)
du_ts = diff(u_ts)
u = u_ts[2:length(u_ts)]
g = g_ts[2:length(u_ts)]
df= data.frame(u, du_ts, g)

# Regress DU_t on present and past growth rates G_{t}, G_{t-1}, G_{t-2}
library(dynlm)
reg.ts = dynlm(d(u_ts)~L(d(u_ts),1)+L(g_ts,0:1))
summary(reg.ts)
```

```
##
## Time series regression with "ts" data:
## Start = 1985(4), End = 2009(3)
##
## Call:
## dynlm(formula = d(u_ts) ~ L(d(u_ts), 1) + L(g_ts, 0:1))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.81304 -0.17489  0.03244  0.13168  1.06162
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.19192    0.04429   4.333 3.75e-05 ***
## L(d(u_ts), 1)  0.39344    0.08270   4.758 7.23e-06 ***
## L(g_ts, 0:1)0 -0.09792    0.02973  -3.294  0.0014 **
## L(g_ts, 0:1)1 -0.13294    0.03100  -4.289 4.43e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2533 on 92 degrees of freedom
## Multiple R-squared:  0.5264, Adjusted R-squared:  0.511
## F-statistic: 34.09 on 3 and 92 DF,  p-value: 6.613e-15
```

```
library(ARDL)
# Estimate the ARDL Model(1,1)
reg.ardl<- ardl(du_ts ~ g , data = df, order = 1)
summary(reg.ardl)
```

```
##
## Time series regression with "ts" data:
## Start = 2, End = 97
```

```
##
## Call:
## dynlm::dynlm(formula = full_formula, data = data, start = start,
##             end = end)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.81304 -0.17489  0.03244  0.13168  1.06162
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.19192    0.04429   4.333 3.75e-05 ***
## L(du_ts, 1)  0.39344    0.08270   4.758 7.23e-06 ***
## g           -0.09792    0.02973  -3.294  0.0014 **
## L(g, 1)      -0.13294    0.03100  -4.289 4.43e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2533 on 92 degrees of freedom
## Multiple R-squared:  0.5264, Adjusted R-squared:  0.511
## F-statistic: 34.09 on 3 and 92 DF,  p-value: 6.613e-15
# Estimate the ARDL Model(2,1)
reg.ardl<- ardl(du_ts ~ g , data = df, order = c(2,1))
summary(reg.ardl)
```

```
##
## Time series regression with "ts" data:
## Start = 3, End = 97
##
## Call:
## dynlm::dynlm(formula = full_formula, data = data, start = start,
##             end = end)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.70751 -0.16250 -0.01653  0.15477  1.03538
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.19089    0.04336   4.403 2.94e-05 ***
## L(du_ts, 1)  0.26015    0.09748   2.669 0.009033 **
## L(du_ts, 2)  0.21955    0.08987   2.443 0.016517 *
## g           -0.10440    0.02922  -3.573 0.000569 ***
## L(g, 1)      -0.12685    0.03053  -4.154 7.41e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2479 on 90 degrees of freedom
## Multiple R-squared:  0.5563, Adjusted R-squared:  0.5365
## F-statistic: 28.21 on 4 and 90 DF,  p-value: 3.434e-15
# Estimate the long-run multipliers of the ARDL model
mult_ardl.lr <- multipliers(reg.ardl, type="lr")
mult_ardl.lr
```

```
##           Term      Estimate Std. Error   t value    Pr(>|t|)
## 1 (Intercept) 0.3668878 0.08360429  4.388385 3.099128e-05
## 2           g -0.4444454 0.08222101 -5.405497 5.257064e-07
```

```
# Estimate the short-run multipliers of the ARDL model
```

```
mult_ardl.sr <- multipliers(reg.ardl, type = "sr")
mult_ardl.sr
```

```
##           Term      Estimate Std. Error   t value    Pr(>|t|)
## 1 (Intercept) 0.1908913 0.04335947  4.402529 2.937483e-05
## 2           g -0.1043979 0.02921813 -3.573053 5.694592e-04
```

```
library(zoo)
```

```
library(xts)
```

```
ce <- coint_eq(reg.ardl, case = 2)
```

```
Okun <- cbind.zoo(du_ts = df[1:97, "du_ts"], ce)
```

```
# Plot the results
```

```
library(xts)
```

```
OkunMult <- xts(Okun, order.by=as.Date(du_ts))
```

```
plot(OkunMult, legend.loc = "left")
```

